LUMENIS / 序境 • SOURCE-AUDITED PRODUCT RECOVERY

# 完整产品架构、功能与十五轮闭环任务书

## 含 44 Screen × 多终端 DESIGN_EXPLORATION 草图（不可直接实现）

不是文件目录汇总，而是从当前源码、调用链、数据库、Permission、Evidence、测试与运行态重新构建的产品地图。

60 CAPABILITIES　40 GAPS　44 SCREENS　225 TASKS　15 CODEX ROUNDS

Baseline: df15a99de6f5ffc3eb85e2c9924afb59c1c3c781 · 2026-09-21

Status: ROUND 01 BASELINE APPROVED · ROUND 02 starts in a new Sol task

EXECUTIVE VERDICT

# 产品没有整体消失，但远未完成

当前最危险的不是代码数量少，而是“存在”被错误写成“可用/已绑定/已完成”。

### Product Skill Store

P0 RESTORE

后端元数据、SQLite、Package Runtime、Execution 与 107 tests 存在；AI-04 却使用旧 FUNCTION_MAP，产品入口和完整生命周期缺失。

### 44 Screen Truth

P0 CORRECT

44 contracts / 54 bindings / 44 HTTP 200，但只有 6 份 detailed spec 和 6 个 Flutter Screen。bound=true 不是完成证明。

### Commercial Surfaces

P0 BUILD

Public Website 与 Web Workspace 均 ABSENT；Desktop/Mobile 仅有六个业务切片。APP ONLY 不再作为目标。

### Core Owners

PARTIAL

DB direct sqlite 已收敛到 Shared Database，但 Permission/Evidence/Context/route/compatibility 仍需逐 caller 闭合。

### Runtime / CI

LIVE_UNVERIFIED

8201、9223、11434 在运行；当前 HEAD 的 Product/App Gate 未运行，最近尝试在 steps 前失败。

### Execution Contract

APPROVED

225 个原 Task 全覆盖到 15 个顺序 Codex 任务。基线已批准；本任务不启动 ROUND 02，也不派发序境09。

ROUND 01 • THREE-WAY TRUTH

# 远程、本地、PDF 三方对照

当前可执行源码和运行证据高于 PDF 与历史说明；PDF 是任务资产，不是完成证明。

| Source | Identity | State | Gap | Disposition |
|---|---|---|---|---|
| Remote | df15a99de6f5 | main | Current HEAD gates NOT RUN | Authority baseline |
| Local | df15a99de6f5 | tracked clean + requested outputs | schedule_state tracked by auto-watch | PROTECT / REVIEW |
| Runtime | 8201 / 9223 / 11434 | Flask / CDP / Ollama | business E2E incomplete | LIVE_UNVERIFIED |
| Original PDF | 235 pages | 225 tasks | App-first scope | fully mapped + multi-surface delta |
| Previous PDF | 82 pages | text descriptions | no 44-screen drawings | replaced by this source-audited edition |

**Immutable boundary**

READ-ONLY

本 PDF 不覆盖原任务书，不改源码，不提交、不 push。所有实现前必须重新 resolve origin/main、HEAD、worktree、owner/callers 与运行证据。

ONE PRODUCT • MULTIPLE SURFACES

# 推荐并已获批准的产品架构

Public Website 与 Web Workspace 分离职责；Desktop 保留本地高级执行；Mobile 是 Companion。

### PUBLIC WEBSITE

产品/能力/行业/案例/Demo/试用/采购/帮助/登录入口

### WEB WORKSPACE

浏览器主操作、真实上下文、任务/结果/Evidence/Decision/Report

### DESKTOP POWER CLIENT

Collector/CDP/Cookie/文件/Ollama/长任务/内网/私有部署

### MOBILE COMPANION

通知、结果、审批、状态、报告、快速处理

### ONE API / VIEWMODEL CONTRACT

SHARED

统一状态语义、Subject / BrandFocus、Permission、Evidence、错误、Return 与 Capability Boundary；不复制业务逻辑。

### ONE PRODUCT CORE

CANONICAL

CTH / BMP / AI / Shared / SYSTEM workflows · Canonical Data / Metrics · Agent / Skill / Reporting / Evaluation / Learning

### ONE TRUTH & EFFECT LAYER

ONLY-ONE

Database Repository · Evidence · Permission / Approval / Effect Fence · Runtime · Collector · Browser · AI Gateway

MULTI-SURFACE DECISION

# 技术路线对比

按复用、SEO、浏览器能力、企业权限、维护成本和第二套 Product Truth 风险评估。

| Option | Reuse | SEO | Browser limits | Cost | Risk | Decision |
|---|---|---|---|---|---|---|
| Flutter Web 单体 | 高 | 弱 | Local/CDP 受限 | 中 | 官网体验与首屏成本 | 不作为全部 Web |
| 独立 Web Workspace | 合同复用 | 中 | 需 Remote boundary | 高 | 第二套状态/逻辑 | 不单独采用 |
| 独立 Public + Flutter Web Workspace | 高 | 强 | Desktop 增强 | 中 | 性能/URL/a11y | 推荐 |
| Public / Workspace 分离部署 | Truth 共享 | 最强 | API handoff | 中 | 版本/登录/观测 | 推荐部署边界 |

## Decision

APPROVED DIRECTION

采用混合路线：独立 Public Website + 以现有 Flutter/Product contracts 为优先复用基线的 Web Workspace；组件复用必须通过浏览器性能、URL、可访问性与安全门。Desktop-only 能力只允许授权发起/查看结果，不在浏览器模拟成功。

SOURCE INVENTORY

# 当前源码规模与诚实状态

计数来自当前 Tree 与 AST/contract/runtime scan；不按文件名推断完整度。

| | | | |
|---|---|---|---|
| **2,117** Tracked files | **1,017** Python files | **44** Dart files | **800** Route decorators |
| **147** Static table names | **191** Root Python callers | **44** Screen contracts | **6** Detailed specs |
| **6** Flutter screens | **54** Backend bindings | **44/44** Screen HTTP 200 | **0** Public/Web product surfaces |

**Key reading**

六个已实现切片、四十四个可返回 builder 与五十四个 pending binding 同时存在；因此当前状态是 PARTIAL，而不是 44 Screen 完成。

RECOVERED FIRST-CLASS CAPABILITY

# LUMENIS Product Skill Store

这是本次源码审计确认的关键遗漏：后端资产存在，但产品目录、入口和生命周期没有闭合。

### DISCOVER

SkillRegistry / built-ins / shared catalog

### INSPECT

publisher · license · I/O · risk · permission

### ADMIT

package preview / digest / disabled registration

### ENABLE

granted scope / config / account state

### VERIFIED RUN

real result / Evidence / history / health

### EVOLVE

version / update / disable / uninstall / rollback

**CURRENT DEFECT**

AI-04 builder currently consumes old function_registry.FUNCTION_MAP (19 functions) instead of canonical SkillRegistry / account skill lifecycle. No Flutter/Web/Mobile Skill Store exists. Round 07 converges backend; Round 14 delivers AI-04; Round 15 releases.

EVIDENCE INDEX

# 本次实际执行与未执行

测试通过只说明相应范围；CI、完整 44 屏与跨表面 Release 仍未证明。

| Area | Scope | Result | Boundary | Status |
|---|---|---|---|---|
| Governance | 6 validators | PASS | current checkout | TESTED |
| Skill / Agent / AI Task | 107 tests | PASS | isolated empty SQLite | TESTED |
| Flutter | 79 tests | PASS | six current slices | TESTED |
| Screen Runtime | 44 endpoints | HTTP 200 / bound=true | not business closure | LIVE_UNVERIFIED |
| Ollama | qwen3:1.7b / 4b / bge-m3 | daemon + inference | derivative only | OPERATING |
| OpenCode | zhipu/glm route | auth failed | no credential exposed | BLOCKED |
| GitHub CI | Product/App Gate | current HEAD not run | latest attempt failed pre-step | BLOCKED |

**防伪：fixture ≠ production • HTTP 200 ≠ business success • provider registered ≠ runtime healthy • missing ≠ 0 • model output ≠ Evidence / Permission / Canonical Truth。**

MISSING CAPABILITY REGISTER

# 缺失 / 风险登记 · 01–08

所有项已进入 15 轮；只在所属 Round/Task Card/owner gate 内实施。

| ID | P | Area | Finding | Status | Round | Required change | Acceptance gate |
|---|---|---|---|---|---|---|---|
| MISS-001 | P0 | Skill Store | LUMENIS Product Skill Store is not visible or whole | PARTIAL | R07 | Converge one catalog/lifecycle owner; preserve built-ins/custom/MCP/module behavior; add Web/Desktop/Mobile role | Discover→inspect trust/license/I-O/permission→enable→verified run→history→update/disable/uninstall; honest health |
| MISS-002 | P0 | Experience Truth | All 44 runtime screens report bound=true despite only six implemented screens/specs | PARTIAL | R08 | Make bound/readiness derive from verified owner/scope/action/evidence and actual surface availability | Unimplemented/blocked fields never report ready/bound; runtime and registry agree |
| MISS-003 | P0 | Web Workspace | Commercial browser workspace absent | ABSENT | R08 | Create approved Flutter Web Workspace shell over canonical API/ViewModel after Round 08 gate | Browser login/context/task/result/Evidence works without local-only capability simulation |
| MISS-004 | P0 | Public Website | Commercial website absent | ABSENT | R08 | Implement separate presentation/acquisition surface: product, CTH/BMP/Shared/SYSTEM, cases/demo, trial/contact/help/login | SEO/accessibility/performance + real trial/contact/workspace-entry journey; no production truth |
| MISS-005 | P0 | Permission | Per-screen READ actor/domain/Subject/BrandFocus permission largely missing | PARTIAL | R04 | Bind trusted actor and target scope through existing Permission owner; fail closed | Unauthorized cross-scope reads denied and audited; UI denial distinct |
| MISS-006 | P0 | Evidence | Shared Evidence reads are globally scoped and duplicate stores remain | PARTIAL | R04 | Converge read contracts and actor/domain/object lineage without rewriting source facts | Every claim/action traces to immutable source and Return; no cross-scope leakage |
| MISS-007 | P0 | Product context | Authorized Subject/BrandFocus does not propagate through all workflows | PARTIAL | R04 | One server-authorized context contract shared by Web/Desktop/Mobile | Context selection cannot grant access; task/history/report all retain scope |
| MISS-008 | P0 | Action closure | Most declared UI actions have no verified API/Permission/effect/Evidence contract | PARTIAL | R04 | Resolve one action at a time through existing owner/firewall/fence | Decision→approval→idempotent effect→result→Evidence→rollback |

MISSING CAPABILITY REGISTER

# 缺失 / 风险登记 · 09–16

所有项已进入 15 轮；只在所属 Round/Task Card/owner gate 内实施。

| ID | P | Area | Finding | Status | Round | Required change | Acceptance gate |
|---|---|---|---|---|---|---|---|
| MISS-009 | P0 | CI | No passing Product/App Gate for current HEAD | BLOCKED | R15 | Repair runner/account/infrastructure and execute current-head blocking gates | Current SHA has complete logs and green required jobs; Tier C remains separate |
| MISS-010 | P0 | Release | No clean-clone/cross-platform/Internal Beta evidence | ABSENT | R15 | Build/install/test Windows/macOS/Linux/Android/iOS as applicable with rollback | Release manifest ties source/schema/tests/artifacts/live evidence |
| MISS-011 | P0 | Security | Sensitive default account/recovery bootstrap material is hardcoded in source | BLOCKED | R04 | Remove fixed secret material through approved migration/recovery design without blocking pre-beta | No plaintext/default personal secret; upgrade/migration/recovery tested |
| MISS-012 | P0 | Runtime output | Mutable Collector schedule state is tracked inside the source tree | EXISTS_BUT_UNSCOPED | R05 | Move future runtime state to canonical operational storage; preserve current behavior/data until owner and migration are approved | Read endpoints do not mutate source tree; operational state survives/recoverable without auto-committing runtime changes |
| MISS-013 | P0 | AI Gateway | Provider status endpoints contradict each other | PARTIAL | R07 | One provider state machine and one contract across AI-01/AI-05 | Same check time/provider/model/fallback yields consistent honest state |
| MISS-014 | P0 | AI API | Duplicate /api/ai/tasks ownership | PARTIAL | R07 | Choose canonical route; adapt callers; preserve behavior/tests before removal | Only one effective route owner; OpenAPI/runtime/callers agree |
| MISS-015 | P0 | 44 screen delivery | 38 screens lack detailed specs and Flutter implementation | ABSENT | R09 | Complete purpose-built spec then Web/Desktop/Mobile composition per scheduled round | Each screen passes field/action/state/surface/accessibility/runtime gates |
| MISS-016 | P1 | Desktop capability boundary | Web cannot express local Collector/CDP/Cookie/Local AI task handoff | ABSENT | R08 | Expose truthful capability availability and authorized desktop/remote task handoff | Web never simulates local success; Desktop result returns to Web |

MISSING CAPABILITY REGISTER

# 缺失 / 风险登记 · 17–24

所有项已进入 15 轮；只在所属 Round/Task Card/owner gate 内实施。

| ID | P | Area | Finding | Status | Round | Required change | Acceptance gate |
|---|---|---|---|---|---|---|---|
| MISS-017 | P1 | Mobile Companion | Notification/approval/report/light-action product flow absent | ABSENT | R14 | Implement companion roles over same contracts | Mobile notification→evidence→approve/deny→result→return |
| MISS-018 | P1 | Jev | Approved typed provider has no implementation | ABSENT | R07 | Implement behind one gateway with typed schema, confidence, shadow eval, cost/legal/privacy and rollback | Never writes truth/effects; low-confidence path explicit; benchmark accepted |
| MISS-019 | P1 | Learning | Continuous learning exists but promotion/rollback is not product-closed | PARTIAL | R07 | Close feedback→evaluation→versioned asset→shadow→approval→promotion→observe/rollback | Version and provenance visible; no autonomous truth/permission rewrite |
| MISS-020 | P1 | Reporting | Report/export paths are multiple and user workflow incomplete | PARTIAL | R07 | Converge owner and artifact lifecycle; preserve CTH/BMP semantics | Scoped preview/export/download with definitions, evidence and audit |
| MISS-021 | P1 | Collector live acceptance | XHS/session/CDP presence not full acquisition proof | LIVE_UNVERIFIED | R06 | Run approved live gate without exposing secrets | Source→capture→Evidence→canonical with rate/risk/recovery and truthful no-data |
| MISS-022 | P1 | Database release | Live WAL blocks ordinary protected-seed test copy; backup/migration/rollback not accepted | BLOCKED | R15 | Use approved external consistent snapshot/backup path; never stop or checkpoint silently | Consistent backup, forward migration, rollback and data integrity evidence |
| MISS-023 | P1 | Governance drift | Product positioning still describes app-only endpoints | PARTIAL | R02 | Reconcile in Round 02 without creating second authority | All current authorities and validators state four surfaces consistently |
| MISS-024 | P1 | Launch path drift | Launch plan references broader D:\LUMENIS_REBUILD path | PARTIAL | R02 | Correct authority pointers after approval | All launch/runbooks resolve canonical repo and actual entry point |

MISSING CAPABILITY REGISTER

# 缺失 / 风险登记 · 25–32

所有项已进入 15 轮；只在所属 Round/Task Card/owner gate 内实施。

| ID | P | Area | Finding | Status | Round | Required change | Acceptance gate |
|---|---|---|---|---|---|---|---|
| MISS-025 | P1 | Binding registry freshness | 54-row registry generated from old SHA | PARTIAL | R01 | Re-resolve every affected row when its round begins; do not blanket mark verified | Row includes current SHA/owner/callers/permission/evidence/tests/runtime |
| MISS-026 | P1 | Function map | Legacy function_map.json has 11/30 statically unmatched endpoint claims | PARTIAL | R02 | Verify runtime url map/callers; preserve unique behavior; absorb or label obsolete | Every catalog function resolves to one current callable or explicit blocked/retired record |
| MISS-027 | P1 | CTH external data | Footfall/member sales/MDM relationships unavailable | BLOCKED | R09 | Integrate only approved POS/CRM/MDM contracts; keep NOT_COLLECTED | No missing→0; source/period/owner and permission visible |
| MISS-028 | P1 | CTH closed loop | Case/decision/action/outcome is not executable in surfaces | PARTIAL | R11 | Close source-backed case/risk/decision/action/outcome/report chain | Real permitted action and outcome evaluation with Evidence lineage |
| MISS-029 | P1 | BMP closed loop | Market→creator/content→campaign→evaluation not executable in surfaces | PARTIAL | R12 | Close BrandFocus-scoped workflow | Real plan/content/campaign result/evaluation with source lineage |
| MISS-030 | P1 | Content studio | SH-01 current App is catalog-only, not a creation studio | PARTIAL | R09 | Implement brief/evidence/editor/version/review/approval/artifact path | Real artifact with citations, version, approval and outcome |
| MISS-031 | P1 | Podcast | Backend/store exists but no product studio | PARTIAL | R13 | Implement audio-first source/timecode/version workflow | Rights/source review and versioned export accepted |
| MISS-032 | P1 | Internal browser | SYS-06 product capture workspace absent | PARTIAL | R14 | Implement controlled evidence capture, not personal browser | Allowed navigation→capture→Evidence→audit/return |

MISSING CAPABILITY REGISTER

# 缺失 / 风险登记 · 33–40

所有项已进入 15 轮；只在所属 Round/Task Card/owner gate 内实施。

| ID | P | Area | Finding | Status | Round | Required change | Acceptance gate |
|---|---|---|---|---|---|---|---|
| MISS-033 | P1 | Notifications/tasks | SH-06 and Mobile task/approval inbox absent | EXISTS_BUT_UNSCOPED | R14 | Add actor-scoped inbox with origin/return | No cross-scope task leakage; actions permissioned and reversible |
| MISS-034 | P1 | Knowledge assets | Resource/intelligence/case libraries lack one current authority and complete entry points | PARTIAL | R11 | Classify operational vs provenance; expose only verified assets | Each asset has owner/version/source/review status and no duplicate truth |
| MISS-035 | P1 | OpenCode engineering | Configured provider fails authentication | BLOCKED | R01 | Repair user-level provider auth outside product source; re-test read-only | Bounded read-only command succeeds without exposing credential |
| MISS-036 | P2 | Accessibility | No complete keyboard/screen-reader/reduced-motion/device matrix | PARTIAL | R15 | Per-screen surface acceptance | WCAG/platform checks plus real device evidence |
| MISS-037 | P2 | Localization | Three-language coverage and long-text layouts incomplete | PARTIAL | R08 | Complete copy ownership and pseudo/long-text acceptance | No truncation/meaning drift across applicable screens/surfaces |
| MISS-038 | P2 | Performance/long-run | No current release-grade long-run evidence | ABSENT | R15 | Measure web/desktop/mobile/runtime/collector/AI under realistic loads | Budgets and recovery accepted for current SHA |
| MISS-039 | P2 | Public commercial funnel | Trial/contact/help/case/demo/enterprise purchase funnel absent | ABSENT | R08 | Define truthful content and CRM/contact handoff without product data ownership | Public→trial/contact/login path works and privacy/analytics are approved |
| MISS-040 | P2 | API/OpenAPI parity | Static route volume and compatibility paths exceed current canonical contract | PARTIAL | R02 | Generate canonical API truth from active app and verify consumers | One operation/owner/schema/error/state per consumer route |

SOURCE-LEVEL CAPABILITY MAP

# 产品能力源码矩阵 · 01–08

Owner → implementation → callers/contract → data → Permission → Evidence → tests → runtime → compatibility。

| ID | Area | Capability | Owner | Implementation | Status | R | Gap |
|---|---|---|---|---|---|---|---|
| CAP-001 | Governance | One Product / Only-One owners | Sol + canonical owner index | Blocking validators and owner index | PARTIAL | R02 | Authority is defined, but current positioning/launch docs still drift. |
| CAP-002 | Surfaces | Public Website | Public presentation surface; no truth owner | No website root, HTML/CSS product surface or deploy config | ABSENT | R08 | Commercial product, demo, trial/contact/help and workspace entry are missing. |
| CAP-003 | Surfaces | Web Workspace | Flutter operational Experience over Product Core | No Flutter web target or browser deployment | ABSENT | R08 | Commercial primary browser workspace is missing. |
| CAP-004 | Surfaces | Desktop Power Client | Flutter App + SYSTEM local capabilities | Six Flutter workflow slices; Windows/macOS/Linux targets exist | PARTIAL | R08 | Only six of 44 screens; no release/device evidence. |
| CAP-005 | Surfaces | Mobile Companion | Flutter mobile composition | Responsive code for six slices; Android/iOS targets exist | PARTIAL | R08 | Notification/approval/report companion is not implemented end to end. |
| CAP-006 | Experience | 44 screen contract set | Experience Constitution | 44 contract directories, 132 JSON files; 54 binding rows | PARTIAL | R08 | Only six detailed specs and six Flutter screens; bound flag overstates readiness. |
| CAP-007 | Experience | Purpose-built UI/UX specifications | Experience Constitution + per-screen specs | 44 one-row blueprints; six detailed specs | PARTIAL | R08 | 38 detailed specs and executable sketches are absent from source. |
| CAP-008 | Experience | Localization / accessibility / motion | Experience foundation | i18n deltas, Flutter semantics/reduced motion in current slices | PARTIAL | R08 | No full 44-screen/4-surface accessibility and locale acceptance. |

SOURCE-LEVEL CAPABILITY MAP

# 产品能力源码矩阵 · 09–16

Owner → implementation → callers/contract → data → Permission → Evidence → tests → runtime → compatibility。

| ID | Area | Capability | Owner | Implementation | Status | R | Gap |
|---|---|---|---|---|---|---|---|
| CAP-009 | Core | Account identity and recovery | SYSTEM / Account | v4 account tables/routes, recovery, TOTP, device/member/approval APIs | PARTIAL | R04 | Post-beta product decision deferred; source contains sensitive bootstrap defaults requiring remediation. |
| CAP-010 | Core | CTH Subject Context | CTH Product Core | subject_context.py + subject_registry.py + runtime identity | PARTIAL | R04 | Read permission and external MDM/association data remain incomplete. |
| CAP-011 | Core | BMP BrandFocus | BMP Product Core | domain context/brand sources and BMP builders | PARTIAL | R04 | Authorized BrandFocus propagation and live workspace selection are incomplete. |
| CAP-012 | Core | Permission / approval | SYSTEM Security | PermissionGate with operation/risk/role/domain decisions and approvals | PARTIAL | R04 | Optional global Basic auth is not per-screen actor/target permission. |
| CAP-013 | Core | Action Firewall / Effect Fence | SYSTEM Security | validate_action + idempotent staging reservations | PARTIAL | R04 | Most Experience actions are declarations without verified write/effect contracts. |
| CAP-014 | Core | Shared Evidence | Shared Evidence | app_evidence_records + chain/manual/API + replay DLQ | PARTIAL | R04 | Actor/domain/object read scope and owner convergence are incomplete. |
| CAP-015 | Core | Canonical Data | Product Core | canonical_data.py plus evidence-chain canonical table and business/platform schemas | PARTIAL | R03 | Canonical writers/readers and dual representations require closure. |
| CAP-016 | Core | Metrics / Truth Table / ViewModel | Product Core + Shared | canonical_metrics.py, ui_view_model.py, viz providers, screen builders | PARTIAL | R03 | Definitions, missing-state semantics and source timestamps are not consistent across all screens. |

SOURCE-LEVEL CAPABILITY MAP

# 产品能力源码矩阵 · 17–24

Owner → implementation → callers/contract → data → Permission → Evidence → tests → runtime → compatibility。

| ID | Area | Capability | Owner | Implementation | Status | R | Gap |
|---|---|---|---|---|---|---|---|
| CAP-017 | Core | Database repository / migrations | Shared Database | Exclusive direct sqlite owner in connection.py/migrations.py | TESTED | R03 | Production migration/rollback and schema convergence remain Round 15 gates. |
| CAP-018 | Core | API truth / composition | Product Core app.py + registered blueprints | 800 static route decorators; app.py and many compatibility blueprints | PARTIAL | R02 | Route ownership and generated OpenAPI/current runtime parity are not closed. |
| CAP-019 | Runtime | Startup / supervisor / health | SYSTEM Runtime | start_production.ps1, run_supervised.py, /healthz | LIVE_UNVERIFIED | R05 | Health payload is shallow and not full dependency readiness. |
| CAP-020 | Runtime | RuntimeGuardian / queue / DLQ | SYSTEM Runtime | runtime_guardian.py and collector guard queues | PARTIAL | R05 | Recovery and long-run worker behavior need live acceptance. |
| CAP-021 | Runtime | Recovery / self-repair | SYSTEM Runtime | auto_recovery, self_repair, full_self_repair, guardians | PARTIAL | R05 | One recovery owner and tested failure/return path not proved. |
| CAP-022 | Runtime | Observability / logs / traces | SYSTEM Runtime | monitoring/logger/ai traces/guardian metrics | PARTIAL | R05 | Ready state is not backed by end-to-end correlated trace acceptance. |
| CAP-023 | Runtime | Settings / module customization | SYSTEM Settings | module_registry + account nav + settings routes | EXISTS_BUT_UNSCOPED | R14 | Useful module/store controls exist but are not surfaced and persistence ownership is split. |
| CAP-024 | Acquisition | Collector orchestration | SYSTEM Collector | collector runtime, adapters, scheduler, guards | PARTIAL | R06 | One runtime owner is declared; live pipeline and compatibility parity remain open. |

SOURCE-LEVEL CAPABILITY MAP

# 产品能力源码矩阵 · 25–32

Owner → implementation → callers/contract → data → Permission → Evidence → tests → runtime → compatibility。

| ID | Area | Capability | Owner | Implementation | Status | R | Gap |
|---|---|---|---|---|---|---|---|
| CAP-025 | Acquisition | Scheduler / rate limit / risk guard | SYSTEM Collector | policy/decider/planner/gate + risk/resource guards | PARTIAL | R06 | Mutable runtime state is now tracked inside the source tree; operational ownership and read-side-effect risk require resolution. |
| CAP-026 | Acquisition | Multi-platform adapters | SYSTEM Collector | XHS, Weibo, Bilibili, Douyin, channels, public web and global platforms | PARTIAL | R06 | Adapters exist at different maturity; registration is not live capability. |
| CAP-027 | Acquisition | Public HTTP / RSS / Atom / Sitemap | SYSTEM Collector | public_source/public_http/web_rss/web_atom/sitemap | IMPLEMENTED | R06 | No Web Workspace source-management UX and no release live matrix. |
| CAP-028 | Acquisition | XHS / RedNote acquisition | SYSTEM Collector + Browser | xhs client/collector/live engine, cookie bridge, CDP collector, HK policy | LIVE_UNVERIFIED | R06 | Cookie/session, 300012, provenance and no-new-data must pass live gate. |
| CAP-029 | Acquisition | Browser / CDP runtime | SYSTEM Browser | router/policy/contract + CDP/Playwright adapters | LIVE_UNVERIFIED | R06 | Resident browser does not prove safe capture, recovery or return. |
| CAP-030 | Acquisition | Cookie / session / auth recovery | SYSTEM Browser/Collector | cookie vault/bridge, browser session, recovery center | PARTIAL | R06 | Must preserve platform identity separation and prove recovery without exposing secrets. |
| CAP-031 | AI | One AI Gateway | SYSTEM AI Gateway | core/channel/providers with compatibility callers | PARTIAL | R07 | Provider-state semantics and duplicate callers are not converged. |
| CAP-032 | AI | Ollama local AI | SYSTEM AI Gateway provider | core auto-detects Ollama; qwen/bge models installed | OPERATING | R07 | Daemon/model are real; user workflow, fallback and Evidence acceptance remain partial. |

# 产品能力源码矩阵 • 33–40

Owner → implementation → callers/contract → data → Permission → Evidence → tests → runtime → compatibility。

| ID | Area | Capability | Owner | Implementation | Status | R | Gap |
|---|---|---|---|---|---|---|---|
| CAP-033 | AI | Zhipu cloud provider | SYSTEM AI Gateway provider | Gateway code/env contract exists | LIVE_UNVERIFIED | R07 | Current live status, cost/privacy/fallback and product use not verified; no credential is included. |
| CAP-034 | AI | Jev typed decision/evaluation provider | SYSTEM AI Gateway candidate | Governance-approved boundary only; no adapter | ABSENT | R07 | Implementation, legal/cost/privacy, shadow eval, confidence handling and rollback absent. |
| CAP-035 | AI | AI task engine | Shared Agent/SYSTEM execution | create/parse/permission/execute/verify/approve/cancel + task/evidence/trace tables | TESTED | R07 | Current API owner and business-scope propagation need convergence. |
| CAP-036 | Agents | Agent runtime / delegation / memory | Shared Agent runtime | adapters, agent groups/capabilities/smart tools/memory | PARTIAL | R07 | Multiple runtime/capability paths and no canonical user workbench closure. |
| CAP-037 | Agents | LUMENIS Product Skill Store | Shared Agent Skill Registry | SkillMetadata registry + SQLite skills_registry + package preview/register/activate + MCP execution/history | PARTIAL | R07 | P0 preserve/restore: backend exists but catalog truth, lifecycle, trust, update/uninstall, health and product UX are split/missing. |
| CAP-038 | Agents | External Skill/MCP package admission | Shared Agent Skill package runtime | GitHub/local/zip/SKILL.md/MCP metadata preview and bounded call | TESTED | R07 | Publisher/license/signature/scanning/update/uninstall and runtime isolation gates incomplete. |
| CAP-039 | Agents | Function and module catalogs | Compatibility callers to canonical registries | FUNCTION_MAP/VIZ_FUNCTIONS/module registry/SHARED_SKILL_CATALOG | PARTIAL | R07 | Must absorb unique behavior into one capability truth without losing module customization. |
| CAP-040 | Shared | Content creation | Shared Creation | content_generation + shared skill catalog + artifact flow | PARTIAL | R07 | No complete editor/version/review/publish workflow. |

SOURCE-LEVEL CAPABILITY MAP

# 产品能力源码矩阵 · 41–48

Owner → implementation → callers/contract → data → Permission → Evidence → tests → runtime → compatibility。

| ID | Area | Capability | Owner | Implementation | Status | R | Gap |
|---|---|---|---|---|---|---|---|
| CAP-041 | Shared | Podcast production | BMP/Shared | podcast_routes + podcast_store + shared skill | PARTIAL | R13 | Audio/transcript/source/version/export workflow absent in surfaces. |
| CAP-042 | Shared | Reporting / export | Shared Reporting | engine/routes/export/artifact/word_pdf | PARTIAL | R07 | One report owner, scoped export, audited artifact retrieval and formats not closed. |
| CAP-043 | Shared | Evaluation | Shared Evaluation | runner/dimensions/recorder/trajectory/benchmark | PARTIAL | R07 | Promotion and cross-domain evaluation semantics need closure. |
| CAP-044 | Shared | Learning loop | Shared Learning | feedback/collector/loop/memory/operations/persistence | PARTIAL | R07 | Continuous learning exists in code but promotion/rollback/observability UI is incomplete. |
| CAP-045 | Shared | Knowledge / resource / case libraries | Shared + domain owners | resource_library, intelligence_library, case libraries/templates | PARTIAL | R11 | Current vs historical authority and product entry points are not fully classified. |
| CAP-046 | Shared | Notifications / task inbox | SYSTEM TaskEngine + adapters | alert engine, notifier, task queues | EXISTS_BUT_UNSCOPED | R14 | Actor-scoped inbox and mobile notification/approval experience absent. |
| CAP-047 | CTH | Monitoring / events / sentiment / trend | CTH | notes/events/trend/risk/calendar/negative modules and screen builders | PARTIAL | R10 | Real source lanes, freshness, event graph and per-screen E2E not closed. |
| CAP-048 | CTH | Case → Insight → Decision → Action → Outcome | CTH + Shared/SYSTEM effects | case/PR/risk/review/action/effect modules | PARTIAL | R11 | Core commercial closed loop not proven end to end. |

SOURCE-LEVEL CAPABILITY MAP

# 产品能力源码矩阵 • 49–56

Owner → implementation → callers/contract → data → Permission → Evidence → tests → runtime → compatibility。

| ID | Area | Capability | Owner | Implementation | Status | R | Gap |
|---|---|---|---|---|---|---|---|
| CAP-049 | CTH | CTH reports / weekly brief | CTH + Shared Reporting | daily/weekly/report engine + evidence exporter | PARTIAL | R11 | Artifact preview/export/retrieval and source trace incomplete. |
| CAP-050 | BMP | Portfolio / competitors / market intelligence | BMP | brand directories/knowledge/market discovery and builders | PARTIAL | R12 | Authorized focus, comparable metrics and market evidence not closed. |
| CAP-051 | BMP | Propagation / campaign | BMP | brand_propagation + topic_campaign/snapshot | PARTIAL | R12 | Metric definitions, windows, actual-vs-goal and outcome closure incomplete. |
| CAP-052 | BMP | Creator ecosystem / KOS/KOL/KOC/UGC | BMP | kos engine/attribution/metrics/paid/routes | PARTIAL | R12 | Identity, authenticity, paid effects and dossier UX need convergence. |
| CAP-053 | BMP | Content lifecycle | BMP + Shared Creation | content generation/center modules | PARTIAL | R13 | Product workflow and rights/effect boundary not surfaced. |
| CAP-054 | BMP | Evaluation / risk / benchmark | BMP + Shared Evaluation | brand evaluation/risk/benchmark builders and functions | PARTIAL | R13 | Comparable baseline, confidence and evidence drill-down absent. |
| CAP-055 | Delivery | Security / compliance | SYSTEM Security | security scanner/boundary/compliance/permission/effect | PARTIAL | R15 | Credential bootstrap, external packages and full release security acceptance remain. |
| CAP-056 | Delivery | CI test tiers | SYSTEM Release | Tier A/B/C manifests and workflows | BLOCKED | R15 | No green current-head CI or Tier C release evidence. |

SOURCE-LEVEL CAPABILITY MAP

# 产品能力源码矩阵 · 57–60

Owner → implementation → callers/contract → data → Permission → Evidence → tests → runtime → compatibility。

| ID | Area | Capability | Owner | Implementation | Status | R | Gap |
|---|---|---|---|---|---|---|---|
| CAP-057 | Delivery | Packaging / clean clone / rollback | SYSTEM Release | deployment/scripts/targets exist | ABSENT | R15 | Windows/macOS/Linux/Android/iOS build and clean-clone evidence missing. |
| CAP-058 | Delivery | Compatibility caller convergence | Per canonical owner | 191 root Python callers classified historically | PARTIAL | R02 | Current HEAD has 191 root callers; unique behavior must be preserved before removal. |
| CAP-059 | Delivery | External business data dependencies | CTH/BMP owners | Explicit not-collected fields and blocked bindings | BLOCKED | R09 | Several promised business outcomes cannot be real until approved external sources exist. |
| CAP-060 | Delivery | Engineering AI tools | Codex engineering layer | Codex active; OpenCode configured; Ollama available | PARTIAL | R01 | OpenCode is currently unusable due to authentication failure; this is an engineering config issue, not product capability. |

FIFTEEN SEQUENTIAL CODEX TASKS

# 十五轮主表 • ROUND 01–05

一轮 = 一个独立 Codex 任务；本任务验收并停止后，用户才决定下一任务。

| | | | |
|---|---|---|---|
| R01 | **Current Truth / PDF Full Coverage**<br>Freeze current remote/local/runtime/PDF truth and protect unknown work. | PDF p.11–14 • 4 tasks | Remote/local/PDF/source/runtime evidence agree; 235 pages and 225 tasks mapped; no source mutation. |
| R02 | **Authority / Only-One / Compatibility**<br>One current product authority, four approved surfaces, named owner for every caller/capability. | PDF p.15–16, 30 • 3 tasks | No App-only drift, duplicate authority or unowned compatibility deletion; Skill Store preserved as a name d capability. |
| R03 | **Database / Canonical Truth**<br>One DB path/repository/schema truth with isolated tests and protected production data. | PDF p.17 • 1 tasks | Canonical paths/writers and rollback proof; no production DB mutation by tests. |
| R04 | **Identity / Context / Permission / Evidence**<br>Trusted actor + Subject/BrandFocus + scoped reads/writes/effects + lineage. | PDF p.18–22 • 5 tasks | Context never grants authorization; every decision/action traces to Evidence and Return. |
| R05 | **Runtime / Health / Recovery / Observability**<br>Honest startup, dependency health, worker/queue/DLQ, recovery and trace. | PDF p.27–28 • 2 tasks | Failure→diagnosis→authorized recovery→Return passes with no source-tree runtime writes. |

FIFTEEN SEQUENTIAL CODEX TASKS

# 十五轮主表 • ROUND 06–10

一轮 = 一个独立 Codex 任务；本任务验收并停止后，用户才决定下一任务。

| | | | |
|---|---|---|---|
| R06 | **Collector / Browser / Acquisition** | One authorized acquisition chain from source/session to Evidence and Canonical Data. | **PDF p.25–26 • 2 tasks**<br>Rate/session/provenance/recovery pass; no second collector/browser. |
| R07 | **AI / Agent / Skill Store / Shared Engines** | One Gateway and Agent runtime; restore full Skill Store; reporting/evaluation/learning remain derivative and governed. | **PDF p.23–24 • 2 tasks**<br>Provider state honest; Skill lifecycle verified; Jev staged; AI never owns truth/effects. |
| R08 | **Cross-Surface Experience Foundation** | Public Website + Web Workspace + Desktop + Mobile roles over one Product Core. | **PDF p.31–34 • 4 tasks**<br>Four distinct surfaces share contract/state/Permission/Evidence; local-only boundaries explicit; no template product. |
| R09 | **Golden Pages + CTH-02** | AI-01, CTH-01/02, BMP-01, SH-01, SYS-01 become real cross-surface workflows. | **PDF p.35, 52–59, 92–95, 136–139, 156–159, 188–191 • 25 tasks**<br>Applicable Web/Desktop/Mobile flow returns real scoped data/result/Evidence; external gaps remain honest. |
| R10 | **CTH Monitoring / Analysis** | CTH-03..07 source-backed live monitoring, event relations, calendar, feedback and trends. | **PDF p.60–79 • 20 tasks**<br>Subject-scoped sources, freshness, Evidence Trace and responsive acceptance pass. |

FIFTEEN SEQUENTIAL CODEX TASKS

# 十五轮主表 · ROUND 11–15

一轮 = 一个独立 Codex 任务；本任务验收并停止后，用户才决定下一任务。

| | | | |
|---|---|---|---|
| R11 | **CTH Case / Risk / Decision / Report** | CTH-08..10 close Case→Insight→Decision→Action→Outcome→Report. | **PDF p.80–91 • 12 tasks**<br>Authorized action/outcome and report lineage are executable and auditable. |
| R12 | **BMP Portfolio / Market / Propagation / Creator** | BMP-02..05 close BrandFocus→market→propagation→creator evidence. | **PDF p.96–111 • 16 tasks**<br>Brand scope, comparable definitions and creator ownership evidence pass. |
| R13 | **BMP Content / Campaign / Podcast / Evaluation** | BMP-06..11 close asset/campaign/podcast/result/evaluation/risk/benchmark. | **PDF p.112–135 • 24 tasks**<br>Rights, approval, effect, actual result and aligned evaluation are traceable. |
| R14 | **Remaining ACC / AI / Shared / SYSTEM** | Deliver ACC-01..04, AI-02..05, SH-02..06, SYS-02..08 with honest deferrals. | **PDF p.36–51, 140–155, 160–187, 192–211 • 80 tasks**<br>Each surface role and state is explicit; account timing honored; no hidden local capability in Web. |
| R15 | **Closed-loop Internal Beta / Clean Clone / Release** | Public→Web→real context/workflow/Evidence→Desktop execution→Web result→Mobile approval→Report/Evaluation/Learning. | **PDF p.29, 212–235 • 25 tasks**<br>Current-SHA cross-surface real-data release evidence and final unresolved list accepted by Sol/PO. |

ROUND 15 TARGET JOURNEY

# 跨表面真实闭环

每一步必须保留 actor、context、Permission、Evidence、state、Return 与错误恢复。

### 01 • Public

了解 / Demo / 试用

### 02 • Web

登录 / Subject / BrandFocus

### 03 • Workflow

任务 / 内容 / 决策

### 04 • Evidence

Trace / Source / Lineage

### 05 • Desktop

CDP / Collector / Local AI

### 06 • Web

结果 / 报告 / Return

### 07 • Mobile

通知 / 审批 / 快速处理

### 08 • Learning

评价 / 版本 / Promotion

### Release Gate

NO FALSE GREEN

真实 permitted data · PC/Mobile/Web · failure/recovery · security · accessibility · performance · Tier A/B/C · clean clone · packages · rollback · final unresolved list。

# PUBLIC-01 • 品牌首页

非设计权威、不可直接实现。商业官网只承载产品叙事、可信 Demo 与转化入口。

**序境 LUMENIS**

产品 · 行业 · 案例 · 帮助 · 登录　申请试用

## 一句话价值与双 CTA

了解产品 → 看 Demo → 申请试用 / 登录

主要行动　查看可信证据

### One Product 闭环图

结构化示意 / 交互式 Demo，不伪造客户或结果。

### One Product 闭环图

清晰定义、真实状态、可回到产品入口。

### CTH/BMP/AI/Shared/SYSTEM 能力索引

清晰定义、真实状态、可回到产品入口。

### Evidence/Permission 信任说明

清晰定义、真实状态、可回到产品入口。

**了解产品 → 看 Demo → 申请试用 / 登录**

Public 不读取生产数据；所有工作区结果由 ONE Product Core 提供。

# PUBLIC-02 • 产品与能力

非设计权威、不可直接实现。商业官网只承载产品叙事、可信 Demo 与转化入口。

**序境 LUMENIS**

产品 · 行业 · 案例 · 帮助 · 登录　申请试用

### 产品域地图

能力 → 适用场景 → Workspace。首屏说明真实适用范围与不可用条件。

### 按角色/问题筛选

清晰定义、真实状态、可回到产品入口。

### 真实 Demo 状态

清晰定义、真实状态、可回到产品入口。

### Desktop-only 能力边界

清晰定义、真实状态、可回到产品入口。

### 能力 → 适用场景 → Workspace

Public 不读取生产数据；所有工作区结果由 ONE Product Core 提供。

# PUBLIC-03 • 行业与使用场景

非设计权威、不可直接实现。商业官网只承载产品叙事、可信 Demo 与转化入口。

**序境 LUMENIS**

产品 · 行业 · 案例 · 帮助 · 登录　申请试用

### 行业选择器

问题 → 工作流 → 预期结果。首屏说明真实适用范围与不可用条件。

### 角色旅程

清晰定义、真实状态、可回到产品入口。

### 业务结果

清晰定义、真实状态、可回到产品入口。

### 数据/权限前置

清晰定义、真实状态、可回到产品入口。

**问题 → 工作流 → 预期结果**

Public 不读取生产数据；所有工作区结果由 ONE Product Core 提供。

# PUBLIC-04 • 客户案例与 Demo

非设计权威、不可直接实现。商业官网只承载产品叙事、可信 Demo 与转化入口。



# PUBLIC-05 • 试用 / 企业采购 / 联系

非设计权威、不可直接实现。商业官网只承载产品叙事、可信 Demo 与转化入口。

**序境 LUMENIS**

产品 · 行业 · 案例 · 帮助 · 登录　申请试用

### 试用路径

资格确认 → 申请 → 跟进。首屏说明真实适用范围与不可用条件。

### 部署/安全选项

清晰定义、真实状态、可回到产品入口。

### 采购资料包

清晰定义、真实状态、可回到产品入口。

### 联系人与 SLA

清晰定义、真实状态、可回到产品入口。

**资格确认 → 申请 → 跟进**

Public 不读取生产数据；所有工作区结果由 ONE Product Core 提供。

# PUBLIC-06 • 帮助 / 登录 / 工作区入口

非设计权威、不可直接实现。商业官网只承载产品叙事、可信 Demo 与转化入口。

**序境 LUMENIS**

产品 · 行业 · 案例 · 帮助 · 登录　申请试用

### 帮助搜索

解决问题 → 进入正确表面。首屏说明真实适用范围与不可用条件。

### 产品状态

清晰定义、真实状态、可回到产品入口。

### 登录/工作区

清晰定义、真实状态、可回到产品入口。

### 桌面客户端下载

清晰定义、真实状态、可回到产品入口。

**解决问题 → 进入正确表面**

Public 不读取生产数据；所有工作区结果由 ONE Product Core 提供。

DESIGN_EXPLORATION • ROUND 14

# ACC-01 • 登录与可信入口

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER BLOCKED　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Enter credentials or select a supported sign-in method; show validation, pending, rejected and recovered states distinctly.

**Responsive recomposition**

Centered form with help/status pane → single-column form with touch-safe fields and explicit return.

**Surface boundary**

Web: 商业入口；pre-beta 不设产品账号墙 · Desktop: 企业身份与私有部署入口 · Mobile: 后 beta 登录/恢复；pre-beta 不阻断

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-025, LUNA-026, LUNA-027, LUNA-028, LUNA-219

DESIGN_EXPLORATION • ROUND 14

# ACC-02 • Subject / Lens 选择

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER BLOCKED　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Search/select an authorized Subject, then confirm an available lens; server response controls the destination.

**Responsive recomposition**

Searchable split list/detail → searchable full-screen list then focused subject confirmation.

**Surface boundary**

Web: 主入口（受产品账号时序约束） · Desktop: 完整身份/恢复能力 · Mobile: 查看、恢复、轻量确认

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-029, LUNA-030, LUNA-031, LUNA-032

DESIGN_EXPLORATION • ROUND 14

# ACC-03 • 工作区启动台

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER BLOCKED　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Resume a restorable work context or start a named product task.

**Responsive recomposition**

Persistent domain rail plus recent-work canvas → compact destinations and one prominent resume/start action.

**Surface boundary**

Web: 主入口（受产品账号时序约束） · Desktop: 完整身份/恢复能力 · Mobile: 查看、恢复、轻量确认

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-033, LUNA-034, LUNA-035, LUNA-036

DESIGN_EXPLORATION • ROUND 14

# ACC-04 • 访问恢复与防护

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER BLOCKED　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Retry, switch to authorized account/context, or request access through a real flow once verified.

**Responsive recomposition**

Diagnostic steps beside the attempted destination → sequential recovery cards with clear back path.

**Surface boundary**

Web: 主入口（受产品账号时序约束） · Desktop: 完整身份/恢复能力 · Mobile: 查看、恢复、轻量确认

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-037, LUNA-038, LUNA-039, LUNA-040

DESIGN_EXPLORATION • ROUND 09

# CTH-01 • 商业运营总览

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC YES　FLUTTER PARTIAL　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Scrub a change on the timeline, inspect its source, open/create a case or decision and trace the follow-up.

**Responsive recomposition**

Parallel metrics + timeline + action queue → one active signal with swipeable evidence and a contextual action sheet.

**Surface boundary**

Web: 主要商业运营入口 · Desktop: 实时采集/深度分析/本地执行增强 · Mobile: 事件、结果、通知、审批

**Acceptance truth**

Implemented Flutter slice, but full action/runtime/device acceptance remains open. PDF tasks: SOL-041, LUNA-042, LUNA-043, LUNA-044

DESIGN_EXPLORATION • ROUND 09

# CTH-02 • 主体 / 商业单元档案

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC YES　FLUTTER PARTIAL　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Expand a relationship and inspect its provenance; switch the selected Subject only through the authorized context control.

**Responsive recomposition**

Hierarchical tree and relationship table → compact breadcrumb hierarchy with focused relationship cards.

**Surface boundary**

Web: 主要商业运营入口 · Desktop: 实时采集/深度分析/本地执行增强 · Mobile: 事件、结果、通知、审批

**Acceptance truth**

Implemented Flutter slice, but full action/runtime/device acceptance remains open. PDF tasks: SOL-045, LUNA-046, LUNA-047, LUNA-048

DESIGN_EXPLORATION • ROUND 10

# CTH-03 • 实时监测

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Pause/resume viewing, pin an event, open source evidence; health actions remain permission-gated.

**Responsive recomposition**

Multi-source live board with filters → prioritized single stream and persistent freshness/connection badge.

**Surface boundary**

Web: 主要商业运营入口 · Desktop: 实时采集/深度分析/本地执行增强 · Mobile: 事件、结果、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-049, LUNA-050, LUNA-051, LUNA-052

DESIGN_EXPLORATION • ROUND 10

# CTH-04 • 事件关系图

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Select a node to reveal linked evidence, navigate to its case, then Trace/Return to the same graph position.

**Responsive recomposition**

Interactive graph + synchronized timeline → vertical event path with evidence nodes and saved zoom/focus.

**Surface boundary**

Web: 主要商业运营入口 · Desktop: 实时采集/深度分析/本地执行增强 · Mobile: 事件、结果、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-053, LUNA-054, LUNA-055, LUNA-056

DESIGN_EXPLORATION • ROUND 10

# CTH-05 • 舆情情绪日历

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Choose a day/range to inspect events and sources; switch to list view for precise keyboard access.

**Responsive recomposition**

Month heatmap beside date detail → week/day agenda with density marks and a slide-up evidence list.

**Surface boundary**

Web: 主要商业运营入口 · Desktop: 实时采集/深度分析/本地执行增强 · Mobile: 事件、结果、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-057, LUNA-058, LUNA-059, LUNA-060

DESIGN_EXPLORATION • ROUND 10

# CTH-06 • 负面监测与处置

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Verify original Source, classify or link to a Case, and preserve the source wording and lineage.

**Responsive recomposition**

Dense triage table with source inspector → stacked review cards with filters and full-screen source reader.

**Surface boundary**

Web: 主要商业运营入口 · Desktop: 实时采集/深度分析/本地执行增强 · Mobile: 事件、结果、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-061, LUNA-062, LUNA-063, LUNA-064

# CTH-07 • 趋势分析

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Add/remove a compatible metric or cohort and inspect the contribution/evidence behind a delta.

**Responsive recomposition**

Resizable synchronized charts and formula table → one primary chart, horizontal metric switcher and tappable delta explanation.

**Surface boundary**

Web: 主要商业运营入口 · Desktop: 实时采集/深度分析/本地执行增强 · Mobile: 事件、结果、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-065, LUNA-066, LUNA-067, LUNA-068

# CTH-08 • Case 与知识库

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Move a case through an authorized state transition; promote a reviewed conclusion to knowledge without overwriting facts.

**Responsive recomposition**

Board/list toggle with case timeline inspector → prioritized case inbox and step-by-step timeline.

**Surface boundary**

Web: 主要商业运营入口 · Desktop: 实时采集/深度分析/本地执行增强 · Mobile: 事件、结果、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-069, LUNA-070, LUNA-071, LUNA-072

DESIGN_EXPLORATION • ROUND 11

# CTH-09 • 风险与决策板

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Inspect basis, assign/record a decision, request approval and confirm the resulting action.

**Responsive recomposition**

Risk matrix plus decision ledger → ranked risk feed with focused decision detail and explicit confirmation.

**Surface boundary**

Web: 主要商业运营入口 · Desktop: 实时采集/深度分析/本地执行增强 · Mobile: 事件、结果、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-073, LUNA-074, LUNA-075, LUNA-076

DESIGN_EXPLORATION • ROUND 11

# CTH-10 • 报告与周报

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Select a period and verified blocks, inspect source list, preview and export with a recorded definition/window.

**Responsive recomposition**

Outline + live page preview + evidence rail → report sections list, one-page preview and share/export sheet.

**Surface boundary**

Web: 主要商业运营入口 · Desktop: 实时采集/深度分析/本地执行增强 · Mobile: 事件、结果、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-077, LUNA-078, LUNA-079, LUNA-080

DESIGN_EXPLORATION • ROUND 09

# BMP-01 • 品牌与市场总览

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC YES　FLUTTER PARTIAL　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Focus a brand, inspect a relationship's evidence and jump into its market, propagation or campaign path.

**Responsive recomposition**

Rich network canvas with linked channel and KPI strips → brand pulse summary with navigable relationship carousel and source trace.

**Surface boundary**

Web: 主要品牌市场入口 · Desktop: 创作、文件、长任务、Local AI 增强 · Mobile: 结果、报告、通知、审批

**Acceptance truth**

Implemented Flutter slice, but full action/runtime/device acceptance remains open. PDF tasks: SOL-081, LUNA-082, LUNA-083, LUNA-084

DESIGN_EXPLORATION • ROUND 12

# BMP-02 • 品牌组合与竞品

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Compare or focus one brand; context switch is explicit and retains the previous focus for Return.

**Responsive recomposition**

Multi-brand canvas and comparison columns → searchable portfolio list with focused brand profile and competitor chips.

**Surface boundary**

Web: 主要品牌市场入口 · Desktop: 创作、文件、长任务、Local AI 增强 · Mobile: 结果、报告、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-085, LUNA-086, LUNA-087, LUNA-088

DESIGN_EXPLORATION • ROUND 12

# BMP-03 • 市场情报

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Filter a market/category window, inspect source provenance and mark a lead for review without calling it confirmed.

**Responsive recomposition**

Radar/landscape with evidence dossier → ranked opportunity feed and source-reader drill-in.

**Surface boundary**

Web: 主要品牌市场入口 · Desktop: 创作、文件、长任务、Local AI 增强 · Mobile: 结果、报告、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-089, LUNA-090, LUNA-091, LUNA-092

DESIGN_EXPLORATION • ROUND 12

# BMP-04 • 传播分析

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Trace a campaign/content item through the funnel and inspect each platform's source, metric definition and freshness.

**Responsive recomposition**

Flow visualization plus channel matrix → vertical funnel with channel selector and explicit metric/source details.

**Surface boundary**

Web: 主要品牌市场入口 · Desktop: 创作、文件、长任务、Local AI 增强 · Mobile: 结果、报告、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-093, LUNA-094, LUNA-095, LUNA-096

# BMP-05 • 创作者生态

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Open a creator dossier, compare verified contributions, note collaboration risk or attach to an authorized campaign.

**Responsive recomposition**

Graph plus sortable roster and dossier → creator cards with fit/risk summary and dossier detail.

**Surface boundary**

Web: 主要品牌市场入口 · Desktop: 创作、文件、长任务、Local AI 增强 · Mobile: 结果、报告、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-097, LUNA-098, LUNA-099, LUNA-100

# BMP-06 • 内容中心

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Upload/version content, review lineage and approval, then request an authorized publish handoff.

**Responsive recomposition**

Swimlane board with asset preview inspector → production queue with version stack and bottom approval actions.

**Surface boundary**

Web: 主要品牌市场入口 · Desktop: 创作、文件、长任务、Local AI 增强 · Mobile: 结果、报告、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-101, LUNA-102, LUNA-103, LUNA-104

# BMP-07 • Campaign 工作区

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Build or revise a plan, assign owners/dates, request approval, record delivery and evaluate against its baseline.

**Responsive recomposition**

Timeline and channel allocation side-by-side → guided milestone sequence with compact objective and status header.

**Surface boundary**

Web: 主要品牌市场入口 · Desktop: 创作、文件、长任务、Local AI 增强 · Mobile: 结果、报告、通知、审批

**Acceptance truth**

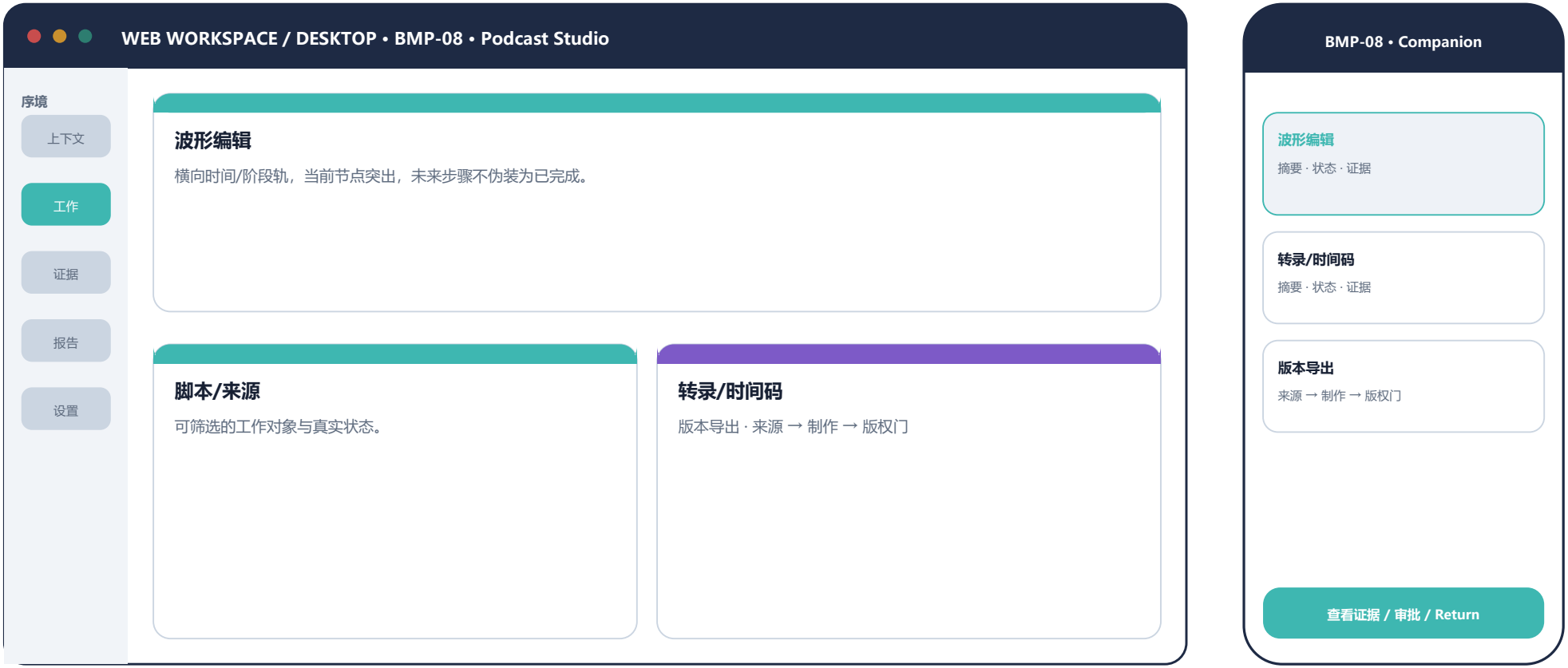
Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-105, LUNA-106, LUNA-107, LUNA-108

DESIGN_EXPLORATION • ROUND 13

# BMP-08 • Podcast Studio

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Draft from an approved brief/evidence set, edit transcript against audio, review source claims and export a versioned cut.

**Responsive recomposition**

Large waveform + script/source side panel → transcript-first editor with scrub controls and a source drawer.

**Surface boundary**

Web: 主要品牌市场入口 · Desktop: 创作、文件、长任务、Local AI 增强 · Mobile: 结果、报告、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-109, LUNA-110, LUNA-111, LUNA-112

# BMP-09 • 品牌评价

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Select a brand/period, inspect a dimension's definition and evidence, then compare only aligned baselines.

**Responsive recomposition**

Score matrix with dimension trend small multiples → prioritized dimension list and one selected evidence-backed trend.

**Surface boundary**

Web: 主要品牌市场入口 · Desktop: 创作、文件、长任务、Local AI 增强 · Mobile: 结果、报告、通知、审批

**Acceptance truth**

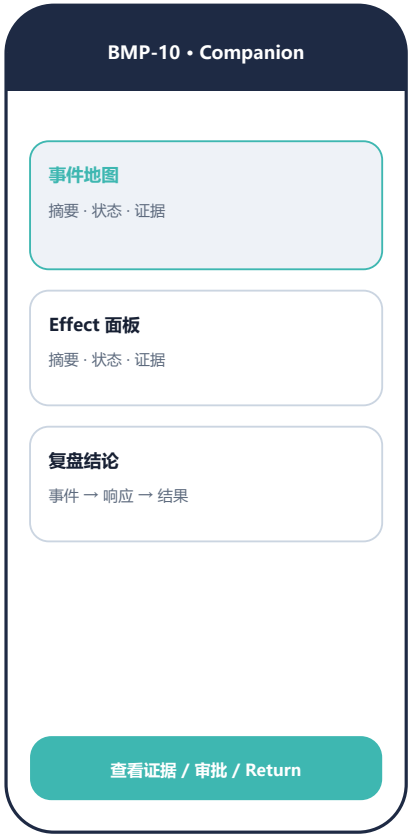
Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-113, LUNA-114, LUNA-115, LUNA-116

# BMP-10 • 品牌风险与事件

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Triage an incident, assign a response, record approval/effect and revisit verified impact.

**Responsive recomposition**

Incident map plus response timeline → urgent queue with a focused incident detail and action ladder.

**Surface boundary**

Web: 主要品牌市场入口 · Desktop: 创作、文件、长任务、Local AI 增强 · Mobile: 结果、报告、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-117, LUNA-118, LUNA-119, LUNA-120

DESIGN_EXPLORATION • ROUND 13

# BMP-11 • Benchmark

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Select comparison set and time window, inspect normalization, identify a supported gap and export a traceable finding.

**Responsive recomposition**

Multi-column benchmark grid and small multiples → swipeable comparison cards with sticky metric definitions.

**Surface boundary**

Web: 主要品牌市场入口 · Desktop: 创作、文件、长任务、Local AI 增强 · Mobile: 结果、报告、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-121, LUNA-122, LUNA-123, LUNA-124

DESIGN_EXPLORATION • ROUND 09

# AI-01 • AI 意图入口

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC YES　FLUTTER PARTIAL　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

State a task, confirm authorized context, inspect evidence/capability plan and start only when an executable route exists.

**Responsive recomposition**

Composer and evidence preview in parallel → single prompt field with context chip and expandable plan preview.

**Surface boundary**

Web: 主要任务与结果入口 · Desktop: 本地 AI、长任务、文件/Browser 增强 · Mobile: 结果、通知、审批、快速追问

**Acceptance truth**

Implemented Flutter slice, but full action/runtime/device acceptance remains open. PDF tasks: SOL-125, LUNA-126, LUNA-127, LUNA-128

DESIGN_EXPLORATION • ROUND 14

# AI-02 • 能力任务工作台

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Start, pause/cancel, retry a safe stage or review a gated action before it runs.

**Responsive recomposition**

Stage graph with artifact inspector → vertical progress path with persistent cancel and expandable evidence.

**Surface boundary**

Web: 主要任务与结果入口 · Desktop: 本地 AI、长任务、文件/Browser 增强 · Mobile: 结果、通知、审批、快速追问

**Acceptance truth**

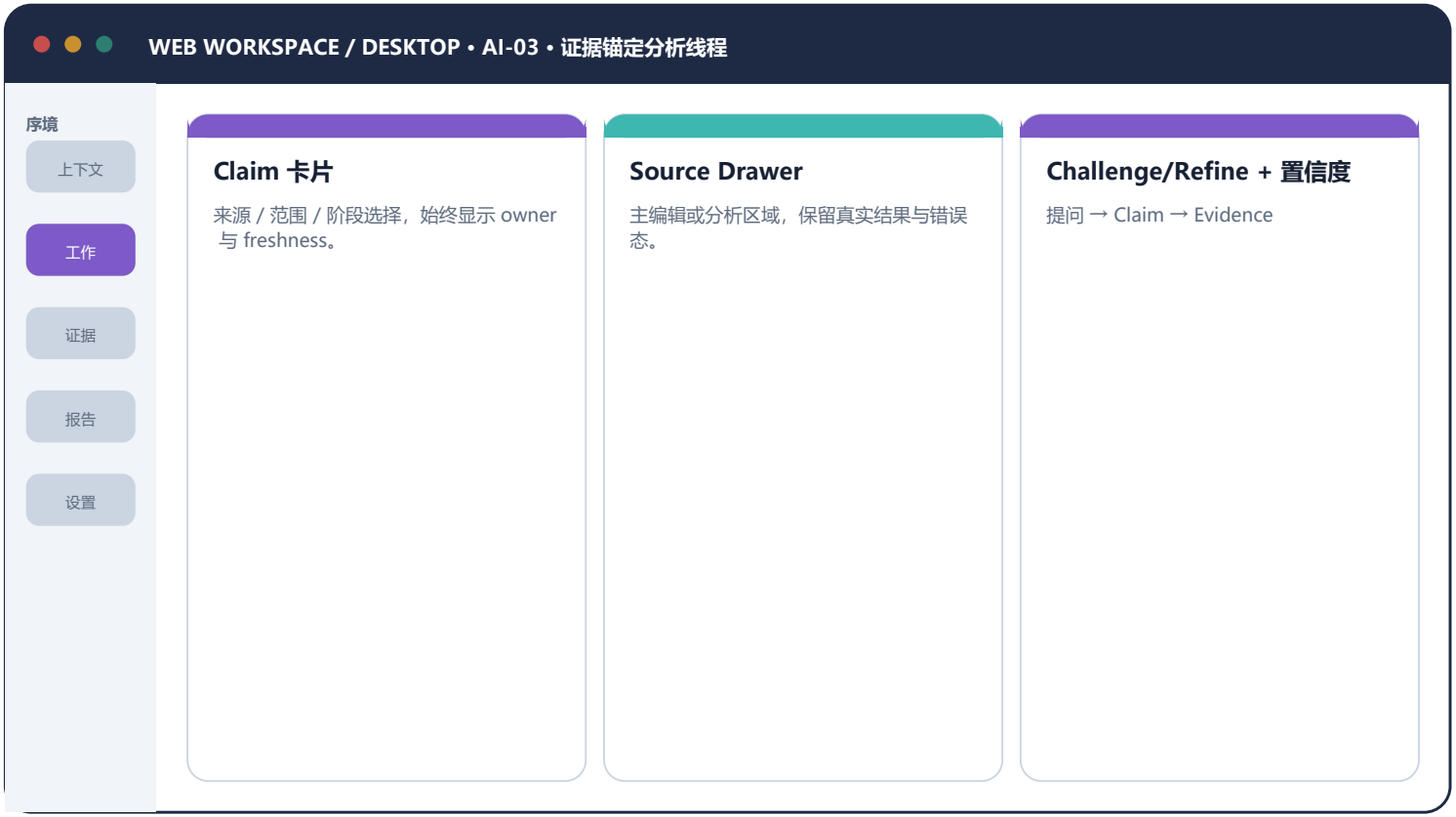
Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-129, LUNA-130, LUNA-131, LUNA-132

DESIGN_EXPLORATION • ROUND 14

# AI-03 • 证据锚定分析线程

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Open a cited source at the relevant passage, challenge/refine a claim or create an owned follow-up task.

**Responsive recomposition**

Thread plus source inspector → claim cards with inline citations and full-screen source return.

**Surface boundary**

Web: 主要任务与结果入口 · Desktop: 本地 AI、长任务、文件/Browser 增强 · Mobile: 结果、通知、审批、快速追问

**Acceptance truth**

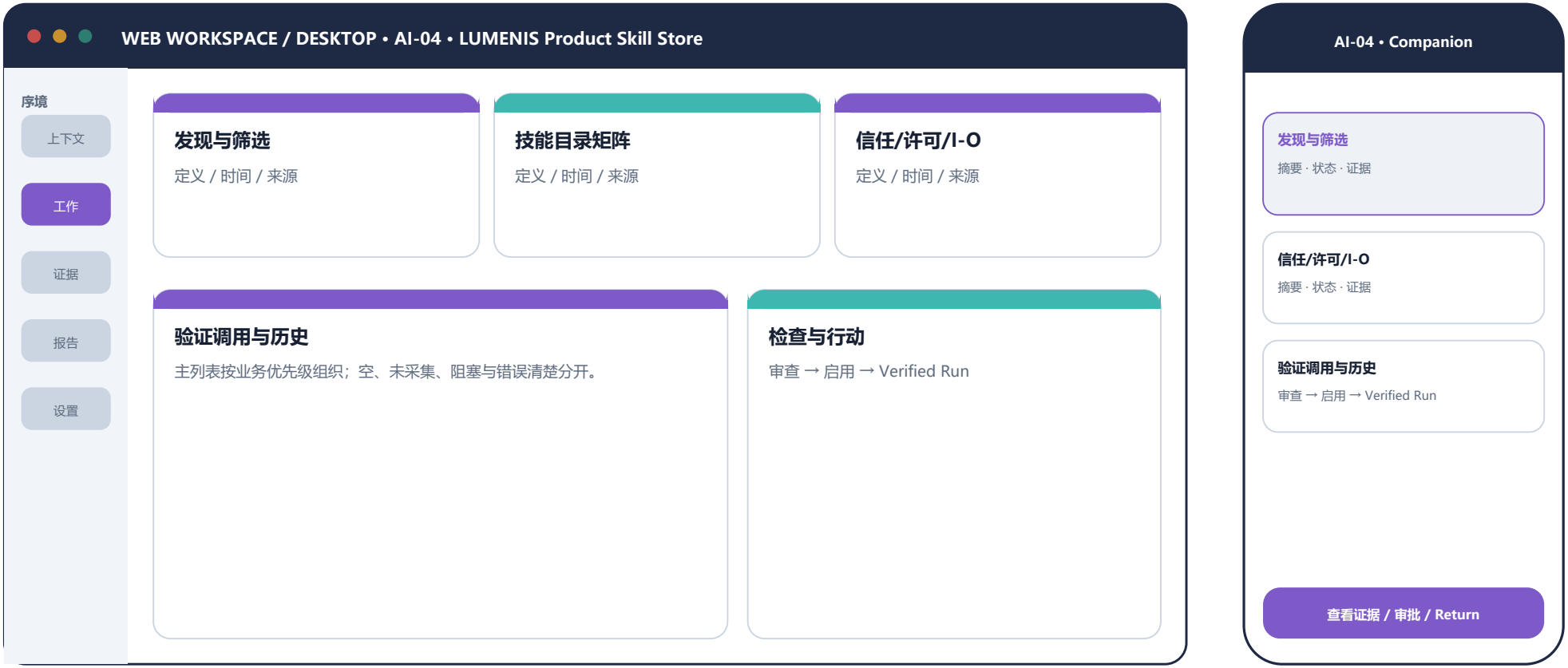
Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-133, LUNA-134, LUNA-135, LUNA-136

DESIGN_EXPLORATION • ROUND 14

# AI-04 • LUMENIS Product Skill Store

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Filter by task, inspect input/output and permission contract, then run a safe verified capability.

**Responsive recomposition**

Matrix/catalog with capability detail panel → searchable capability list and detail sheet.

**Surface boundary**

Web: 主要任务与结果入口 · Desktop: 本地 AI、长任务、文件/Browser 增强 · Mobile: 结果、通知、审批、快速追问

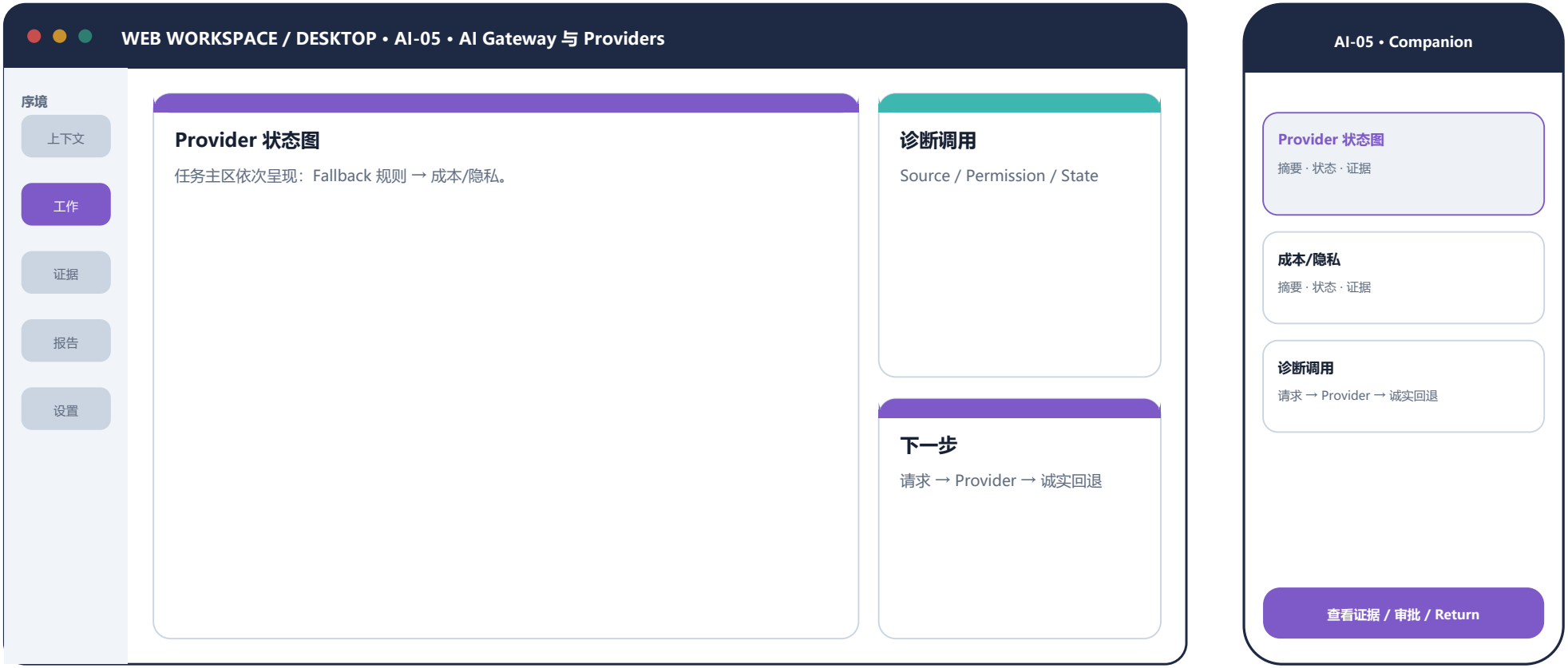
**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-137, LUNA-138, LUNA-139, LUNA-140

# AI-05 • AI Gateway 与 Providers

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Inspect provider state or run a diagnostic; fallback must be observable and must not block core operations.

**Responsive recomposition**

Gateway flow map with health table → provider status list with explicit fallback/last-check details.

**Surface boundary**

Web: 主要任务与结果入口 · Desktop: 本地 AI、长任务、文件/Browser 增强 · Mobile: 结果、通知、审批、快速追问

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-141, LUNA-142, LUNA-143, LUNA-144

DESIGN_EXPLORATION • ROUND 09

# SYS-01 • Runtime Health

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC YES　FLUTTER PARTIAL　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Open a failing dependency's diagnostic, inspect trace and invoke only an authorized recovery.

**Responsive recomposition**

Dependency map with health timeline → ordered dependency list and focused diagnostic view.

**Surface boundary**

Web: 授权运营查看与轻量处置 · Desktop: 深度诊断、恢复与审计 · Mobile: 告警、诊断摘要、审批

**Acceptance truth**

Implemented Flutter slice, but full action/runtime/device acceptance remains open. PDF tasks: SOL-145, LUNA-146, LUNA-147, LUNA-148

DESIGN_EXPLORATION • ROUND 14

# SYS-02 • Collector Status

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Inspect a run, retry/stop when allowed, and trace a collected record to its destination state.

**Responsive recomposition**

Live operations table with per-platform drilldown → platform queue cards with persistent running/retry status.

**Surface boundary**

Web: 远程状态、授权发起、结果查看 · Desktop: 主要/独占本地 Collector、CDP、Cookie、文件能力 · Mobile: 状态、通知、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-149, LUNA-150, LUNA-151, LUNA-152

DESIGN_EXPLORATION • ROUND 14

# SYS-03 • Database Truth

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Inspect a table/chain validation and open exact integrity errors; recovery is a separately authorized path.

**Responsive recomposition**

Schema tree with chain timeline and diagnostics → chain stages with integrity summary and error detail.

**Surface boundary**

Web: 授权运营查看与轻量处置 · Desktop: 深度诊断、恢复与审计 · Mobile: 告警、诊断摘要、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-153, LUNA-154, LUNA-155, LUNA-156

# SYS-04 • Security & Permission

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Simulate/read an access decision, inspect policy reason, or approve only through a real second-factor/approval contract.

**Responsive recomposition**

Matrix and audit ledger → searchable role/scope list with explicit decision detail.

**Surface boundary**

Web: 授权运营查看与轻量处置 · Desktop: 深度诊断、恢复与审计 · Mobile: 告警、诊断摘要、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-157, LUNA-158, LUNA-159, LUNA-160

# SYS-05 • 设置与外观

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Change a local/account preference, preview, save and restore defaults; announce persisted scope.

**Responsive recomposition**

Side-by-side preference controls and live preview → grouped settings pages with immediate accessible preview.

**Surface boundary**

Web: 工作区偏好设置 · Desktop: 本地/企业设置增强 · Mobile: 个人偏好轻配置

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-161, LUNA-162, LUNA-163, LUNA-164

DESIGN_EXPLORATION • ROUND 14

# SYS-06 • Internal Browser

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Navigate only within allowed capture rules, pin a snapshot as Evidence and inspect the audit result.

**Responsive recomposition**

Browser viewport plus evidence inspector → source capture full screen with a compact metadata/trace panel.

**Surface boundary**

Web: 远程状态、授权发起、结果查看 · Desktop: 主要/独占本地 Collector、CDP、Cookie、文件能力 · Mobile: 状态、通知、审批

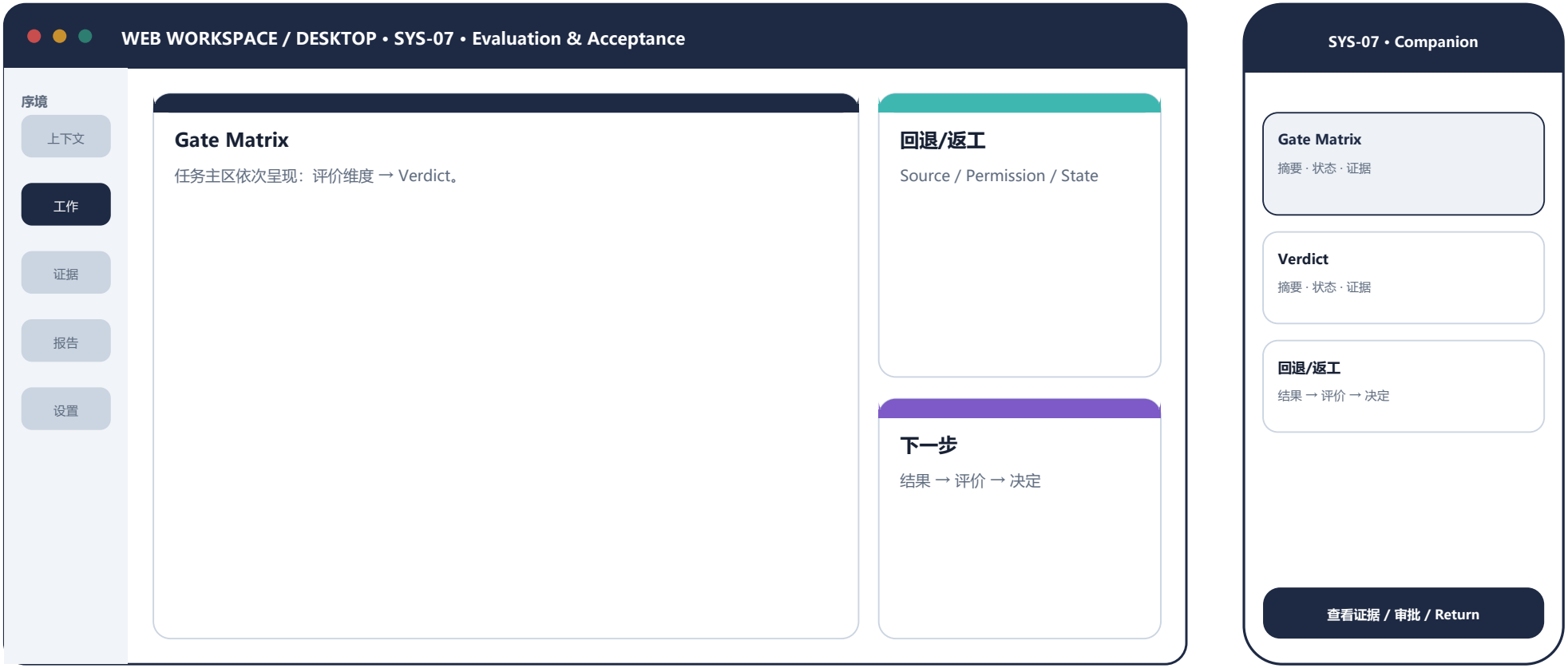
**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-165, LUNA-166, LUNA-167, LUNA-168

# SYS-07 • Evaluation & Acceptance

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Open a gate's artifact, rerun a permitted check and inspect exact failing criterion.

**Responsive recomposition**

Gate matrix with evidence detail → sequential gate cards with current blocker and artifact links.

**Surface boundary**

Web: 授权运营查看与轻量处置 · Desktop: 深度诊断、恢复与审计 · Mobile: 告警、诊断摘要、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-169, LUNA-170, LUNA-171, LUNA-172

# SYS-08 • Observability & Logs

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Filter by trace/time/component, inspect correlated logs and return to the originating runtime or task.

**Responsive recomposition**

Waterfall and query facets with detail pane → trace summary followed by scrollable spans and errors.

**Surface boundary**

Web: 授权运营查看与轻量处置 · Desktop: 深度诊断、恢复与审计 · Mobile: 告警、诊断摘要、审批

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-173, LUNA-174, LUNA-175, LUNA-176

DESIGN_EXPLORATION • ROUND 09

# SH-01 • 内容创作 Studio

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC YES　FLUTTER PARTIAL　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Create a draft from a real brief, cite evidence, review revisions and hand off for human approval.

**Responsive recomposition**

Three-zone canvas (sources/work/artifact) → focused editor with source drawer and version/action sheet.

**Surface boundary**

Web: 主要共享工作流入口 · Desktop: 文件、导出、长任务增强 · Mobile: 阅读、通知、审批、轻操作

**Acceptance truth**

Implemented Flutter slice, but full action/runtime/device acceptance remains open. PDF tasks: SOL-177, LUNA-178, LUNA-179, LUNA-180

DESIGN_EXPLORATION • ROUND 14

# SH-02 • Evidence Library

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Search/filter, compare source with extracted item, inspect fresh ness and open its owning CTH/BMP origin.

**Responsive recomposition**

Dense result list with provenance inspector → searchable evide nce cards with full-screen source trace.

**Surface boundary**

Web: 主要共享工作流入口 · Desktop: 文件、导出、长任务增强 · M obile: 阅读、通知、审批、轻操作

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Ex perience slice or detailed spec. PDF tasks: SOL-181, LUNA-182, LUNA-183, LUNA-184

DESIGN_EXPLORATION • ROUND 14

# SH-03 • Canonical Data Browser

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Inspect a field or metric and traverse lineage; this is read-only exploration, not business fact editing.

**Responsive recomposition**

Navigable schema tree plus query/result panes → table/metric list with definition and lineage drilldown.

**Surface boundary**

Web: 主要共享工作流入口 · Desktop: 文件、导出、长任务增强 · Mobile: 阅读、通知、审批、轻操作

**Acceptance truth**

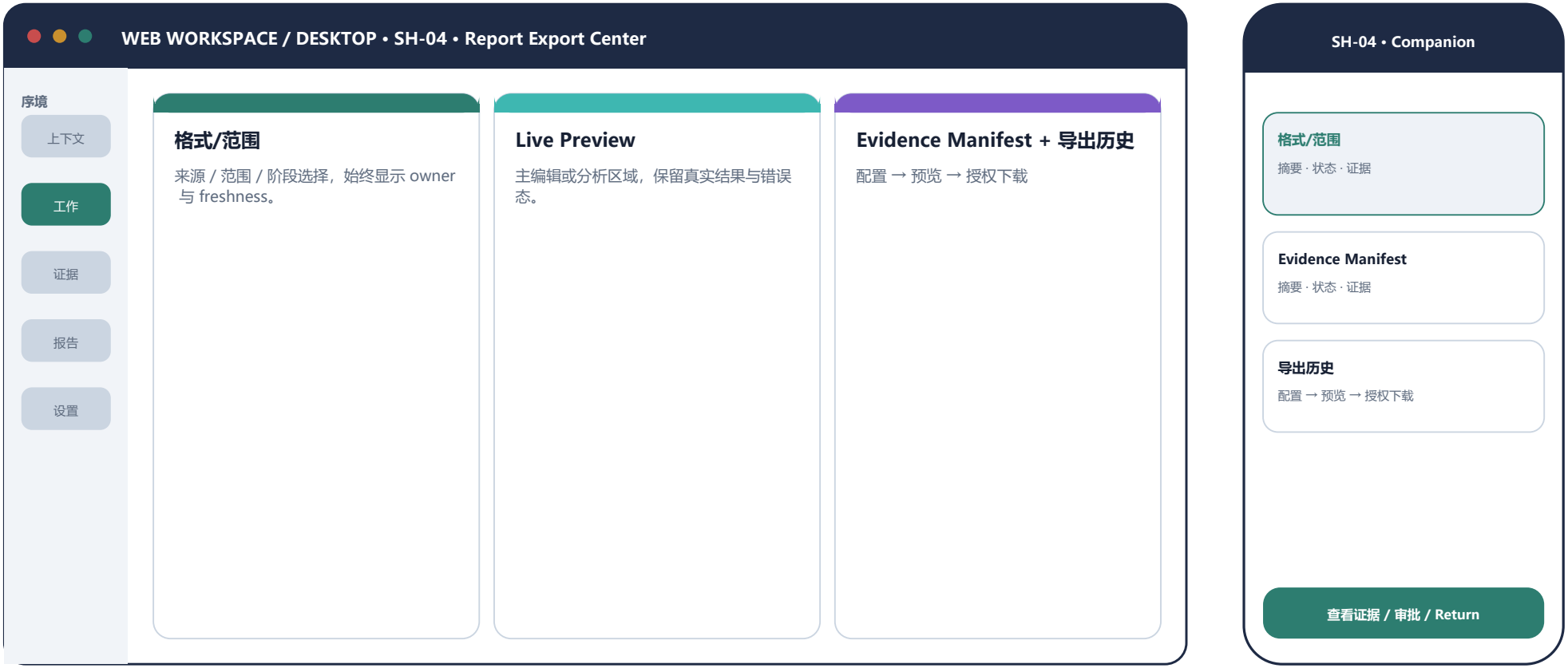
Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-185, LUNA-186, LUNA-187, LUNA-188

DESIGN_EXPLORATION • ROUND 14

# SH-04 • Report Export Center

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True




## PRODUCT CONTRACT

**Primary interaction**

Choose PDF/Excel/PPT only when supported, set scope, inspect preview, export and retrieve audited artifact.

**Responsive recomposition**

Export configuration and artifact preview split → guided export steps and a final downloadable artifact view.

**Surface boundary**

Web: 主要共享工作流入口 · Desktop: 文件、导出、长任务增强 · Mobile: 阅读、通知、审批、轻操作

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-189, LUNA-190, LUNA-191, LUNA-192

DESIGN_EXPLORATION • ROUND 14

# SH-05 • Learning Assets

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Open a learning asset, inspect verification history, propose a revision or use it as a cited task input.

**Responsive recomposition**

Asset shelves with review timeline → filtered learning library and one-asset reading view.

**Surface boundary**

Web: 主要共享工作流入口 · Desktop: 文件、导出、长任务增强 · Mobile: 阅读、通知、审批、轻操作

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-193, LUNA-194, LUNA-195, LUNA-196

DESIGN_EXPLORATION • ROUND 14

# SH-06 • 通知与任务

非设计权威、不可直接实现。须经 Experience Constitution + 详细 Screen Spec + Sol DESIGN_READY 后才可编码。

SPEC NO　FLUTTER ABSENT　RUNTIME HTTP 200 / bound=True







## PRODUCT CONTRACT

**Primary interaction**

Triage, assign/snooze/resolve only through authorized actions; every item returns to its originating task/evidence.

**Responsive recomposition**

Multi-lane inbox with detail inspector → prioritized queue and full-screen task action with clear origin/return.

**Surface boundary**

Web: 主要共享工作流入口 · Desktop: 文件、导出、长任务增强 · Mobile: 阅读、通知、审批、轻操作

**Acceptance truth**

Backend builder/contract may exist, but no canonical Flutter Experience slice or detailed spec. PDF tasks: SOL-197, LUNA-198, LUNA-199, LUNA-200

# Screen 实现与表面状态 · 01–11

runtime bound=true 不提升 UI 或业务状态。

| Screen | Spec | Binding | Flutter | Web | Desktop | Mobile | R | PDF tasks |
|---|---|---|---|---|---|---|---|---|
| ACC-01 | NO | PARTIAL:1 | BLOCKED | ABSENT | ABSENT | ABSENT | R14 | SOL-025, LUNA-026, LUNA-027, LUNA-028, LUNA-219 |
| ACC-02 | NO | PARTIAL:1 | BLOCKED | ABSENT | ABSENT | ABSENT | R14 | SOL-029, LUNA-030, LUNA-031, LUNA-032 |
| ACC-03 | NO | PARTIAL:1 | BLOCKED | ABSENT | ABSENT | ABSENT | R14 | SOL-033, LUNA-034, LUNA-035, LUNA-036 |
| ACC-04 | NO | PARTIAL:1 | BLOCKED | ABSENT | ABSENT | ABSENT | R14 | SOL-037, LUNA-038, LUNA-039, LUNA-040 |
| CTH-01 | YES | PARTIAL:3 | PARTIAL | ABSENT | PARTIAL | PARTIAL | R09 | SOL-041, LUNA-042, LUNA-043, LUNA-044 |
| CTH-02 | YES | PARTIAL:1 | PARTIAL | ABSENT | PARTIAL | PARTIAL | R09 | SOL-045, LUNA-046, LUNA-047, LUNA-048 |
| CTH-03 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R10 | SOL-049, LUNA-050, LUNA-051, LUNA-052 |
| CTH-04 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R10 | SOL-053, LUNA-054, LUNA-055, LUNA-056 |
| CTH-05 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R10 | SOL-057, LUNA-058, LUNA-059, LUNA-060 |
| CTH-06 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R10 | SOL-061, LUNA-062, LUNA-063, LUNA-064 |
| CTH-07 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R10 | SOL-065, LUNA-066, LUNA-067, LUNA-068 |

# Screen 实现与表面状态 · 12–22

runtime bound=true 不提升 UI 或业务状态。

| Screen | Spec | Binding | Flutter | Web | Desktop | Mobile | R | PDF tasks |
|---|---|---|---|---|---|---|---|---|
| CTH-08 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R11 | SOL-069, LUNA-070, LUNA-071, LUNA-072 |
| CTH-09 | NO | ABSENT:1; PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R11 | SOL-073, LUNA-074, LUNA-075, LUNA-076 |
| CTH-10 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R11 | SOL-077, LUNA-078, LUNA-079, LUNA-080 |
| BMP-01 | YES | PARTIAL:1 | PARTIAL | ABSENT | PARTIAL | PARTIAL | R09 | SOL-081, LUNA-082, LUNA-083, LUNA-084 |
| BMP-02 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R12 | SOL-085, LUNA-086, LUNA-087, LUNA-088 |
| BMP-03 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R12 | SOL-089, LUNA-090, LUNA-091, LUNA-092 |
| BMP-04 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R12 | SOL-093, LUNA-094, LUNA-095, LUNA-096 |
| BMP-05 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R12 | SOL-097, LUNA-098, LUNA-099, LUNA-100 |
| BMP-06 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R13 | SOL-101, LUNA-102, LUNA-103, LUNA-104 |
| BMP-07 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R13 | SOL-105, LUNA-106, LUNA-107, LUNA-108 |
| BMP-08 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R13 | SOL-109, LUNA-110, LUNA-111, LUNA-112 |

# Screen 实现与表面状态 · 23–33

runtime bound=true 不提升 UI 或业务状态。

| Screen | Spec | Binding | Flutter | Web | Desktop | Mobile | R | PDF tasks |
|---|---|---|---|---|---|---|---|---|
| BMP-09 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R13 | SOL-113, LUNA-114, LUNA-115, LUNA-116 |
| BMP-10 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R13 | SOL-117, LUNA-118, LUNA-119, LUNA-120 |
| BMP-11 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R13 | SOL-121, LUNA-122, LUNA-123, LUNA-124 |
| AI-01 | YES | PARTIAL:2 | PARTIAL | ABSENT | PARTIAL | PARTIAL | R09 | SOL-125, LUNA-126, LUNA-127, LUNA-128 |
| AI-02 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-129, LUNA-130, LUNA-131, LUNA-132 |
| AI-03 | NO | ABSENT:1; EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-133, LUNA-134, LUNA-135, LUNA-136 |
| AI-04 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-137, LUNA-138, LUNA-139, LUNA-140 |
| AI-05 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-141, LUNA-142, LUNA-143, LUNA-144 |
| SYS-01 | YES | PARTIAL:1 | PARTIAL | ABSENT | PARTIAL | PARTIAL | R09 | SOL-145, LUNA-146, LUNA-147, LUNA-148 |
| SYS-02 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-149, LUNA-150, LUNA-151, LUNA-152 |
| SYS-03 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-153, LUNA-154, LUNA-155, LUNA-156 |

# Screen 实现与表面状态 · 34–44

runtime bound=true 不提升 UI 或业务状态。

| Screen | Spec | Binding | Flutter | Web | Desktop | Mobile | R | PDF tasks |
|---|---|---|---|---|---|---|---|---|
| SYS-04 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-157, LUNA-158, LUNA-159, LUNA-160 |
| SYS-05 | NO | LOCAL_ONLY:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-161, LUNA-162, LUNA-163, LUNA-164 |
| SYS-06 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-165, LUNA-166, LUNA-167, LUNA-168 |
| SYS-07 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-169, LUNA-170, LUNA-171, LUNA-172 |
| SYS-08 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-173, LUNA-174, LUNA-175, LUNA-176 |
| SH-01 | YES | ABSENT:3; PARTIAL:3 | PARTIAL | ABSENT | PARTIAL | PARTIAL | R09 | SOL-177, LUNA-178, LUNA-179, LUNA-180 |
| SH-02 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-181, LUNA-182, LUNA-183, LUNA-184 |
| SH-03 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-185, LUNA-186, LUNA-187, LUNA-188 |
| SH-04 | NO | PARTIAL:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-189, LUNA-190, LUNA-191, LUNA-192 |
| SH-05 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-193, LUNA-194, LUNA-195, LUNA-196 |
| SH-06 | NO | EXISTS_BUT_UNSCOPED:1 | ABSENT | ABSENT | ABSENT | ABSENT | R14 | SOL-197, LUNA-198, LUNA-199, LUNA-200 |

ORIGINAL TASKBOOK • FULL COVERAGE

# PDF Task 映射 • 001–009 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 11 | SOL-000 | SOL | 接管规则 + 审计基线冻结 | R01 | PARTIAL | 把这份任务书作为 Sol 产品经理窗口的唯一执行索引；每次任务前重新 resolve origin/main，不把 PDF 中的 SHA 当未来代码真相。 | Sol 能对每个后续 Task 给出：GO / BLOCKED / SPLIT / REORDER，并要求 Luna 返回 commit + tests + blockers。 |
| 12 | SOL-001 | SOL | 确定 Internal Beta MVP 闭环 | R01 | PARTIAL | 从现有产品真实能力中确定第一版内测最小闭环，必须能从真实 Source/Evidence 走到用户结果与 Return，而不是“一个页面能打开”。 | 输出 3-5 条 beta critical journeys，并给每条明确 entry → action → result → evidence → return/evaluation。 |
| 13 | SOL-002 | SOL | 建立 Batch 依赖图与上下文重置规则 | R01 | PARTIAL | 把全部任务按 Truth owner 和依赖关系排成可并行/不可并行 Batch；明确何时 Luna 继续同一窗口，何时必须新窗口。 | 每个 Batch 有入口 SHA、允许路径、禁止路径、验收点、退出条件；上下文超过一个 owner family 或出现漂移时新开 Luna chat。 |
| 14 | LUNA-003 | LUNA | 现场 Git / Branch / 工作树校验 | R01 | PARTIAL | 在本地执行机确认 main 与 origin/main 一致、无隐藏 side branch、保存未知未提交工作，并记录真实起始 SHA。 | 返回 current HEAD、origin/main、remote heads、git status、是否 clean；任何差异精确列出。 |
| 15 | LUNA-004 | LUNA | 191 个根级 Python Caller 分类清单 | R02 | PARTIAL | 逐个分类根目录 191 个 Python 文件为 canonical owner / active compatibility caller / migrate-ready / provenance-only / delete-ready，禁止凭文件名删。 | 每个根文件有 owner、unique behavior、direct caller、tests、迁移条件；没有证据则 keep/unknown。 |
| 16 | SOL-005 | SOL | 审核 Root Caller 收敛矩阵 | R02 | PARTIAL | 审核 LUNA-004，决定每组 caller 的 keep / adapt / absorb / retire 次序；把 destructive cleanup 推迟到唯一行为完成迁移后。 | 每组兼容层得到明确决策与后续 Luna task；UNKNOWN 不进入删除。 |
| 17 | LUNA-006 | LUNA | Shared Database Authority 最终收敛 | R03 | TESTED | 证明所有生产 DB 访问经过 lumenis/shared/database；兼容 database.py/db_paths.py/migrations wrappers 只保留必要 API，不再形成第二 owner。 | tracked production source 直接 sqlite3.connect=0；path/migration/repository authority 唯一；wrapper 行为回归绿。 |
| 18 | LUNA-007 | LUNA | Pre-beta 无产品账号墙 + Dual Auth 边界 | R04 | PARTIAL | 让 App 在 internal beta 前不被 LUMENIS 产品账号墙阻塞，同时保留 Core Basic/account session 与服务端 Permission；媒体 Cookie 身份完全分离。 | App 可进入 pre-beta shell；受保护操作仍真实 401/403/Permission；双通道状态文案准确。 |
| 19 | LUNA-008 | LUNA | CTH Subject Context + per-Subject Read Permission | R04 | PARTIAL | 把 authenticated actor、CTH Subject context、逐 Subject 业务 read Permission 串成可信服务端决策，供所有 CTH screens 复用。 | 服务端能返回/执行 actor→Subject read decision；denied/unknown fail closed；所有 CTH screen 可复用同一 contract。 |

# PDF Task 映射 • 010–018 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 20 | LUNA-009 | LUNA | BMP BrandFocus Context + Authorization | R04 | PARTIAL | 建立/收敛现有 BMP BrandFocus 的可信 actor scope 与 read authorization，替代产品级 ZBMP 聚合充当选中品牌。 | BrandFocus 由可信服务端 context 决定；授权/未授权/无 focus 分状态；BMP 查询能按 focus scope。 |
| 21 | LUNA-010 | LUNA | Shared Evidence Actor/Domain/Object Scope | R04 | PARTIAL | 给 Shared Evidence 读路径补 actor + product/domain + object/origin scope，并保留 source identity/freshness/lineage。 | 可按 CTH Subject/BMP BrandFocus/AI execution/origin 查询；无 scope fail closed；原始 Evidence 不被 AI/human output 覆盖。 |
| 22 | LUNA-011 | LUNA | Reporting / Export Permission + Lineage | R04 | PARTIAL | 让 report/export 从同一 canonical truth + Evidence lineage 生成，并通过 Permission/approval/effect fence；失败不可假成功。 | 报告 preview/export 同源；scope/period/methodology/Evidence 可追；export audit 完整。 |
| 23 | LUNA-012 | LUNA | AI Gateway 单一 Runtime Owner | R07 | PARTIAL | 确认所有产品 AI caller 收敛到 lumenis/system/ai_gateway，local_ai_client.py 只保留兼容 adapter 或吸收后 retire；Ollama provider live state 真实可见。 | 单一 provider/fallback status contract；真实 probe 区分 configured/ready/degraded/offline；兼容 caller 有清晰去向。 |
| 24 | LUNA-013 | LUNA | AI Task / Execution Route 单一化 | R07 | PARTIAL | 收敛 ai_api.py / ai_task_engine / shared agents skills 的 task/execution route，消除重复声明/平行执行语义。 | 一个 canonical execution path；create/run/cancel/evidence/approval 语义一致；重复 route 消失或明确 compatibility。 |
| 25 | LUNA-014 | LUNA | SYSTEM Collector 单一 Runtime 收敛 | R06 | PARTIAL | 把 root collector/automation callers 绑定到 lumenis/system/collector；各平台 adapter 统一 Source→Evidence→Canonical，保留 Cookie/session 策略。 | collector registry/runtime/scheduler/guard 的 owner 唯一；root caller 仅 compatibility；关键平台回归绿。 |
| 26 | LUNA-015 | LUNA | Browser/CDP + Media Cookie Session 收敛 | R06 | PARTIAL | 统一 Browser Surface / browser runtime / XHS Cookie bridge 调用边界，保持 9223 HK collector 与 9224 general browser 的真实用途和 provenance。 | 浏览/采集 session purpose 可辨；policy/allowed target/provenance 生效；无 secret 泄漏。 |
| 27 | LUNA-016 | LUNA | RuntimeGuardian / Recovery Owner 收敛 | R05 | PARTIAL | RuntimeGuardian、supervisor、recovery 仅表达 SYSTEM runtime truth；DB access 走 repository，状态不与业务健康混淆。 | 健康、degraded、offline、unknown、recovery 明确；单 supervisor 逻辑可重复启动。 |
| 28 | LUNA-017 | LUNA | 启动链与本地 Production Preflight | R05 | PARTIAL | 验证 canonical Windows launcher 从依赖→supervisor→8201 health→9223/9224 browser 初始化到关键 API，失败能定位。 | 重复启动幂等；8201 唯一 supervisor；healthz 与关键 API 可用；browser health 分开报告。 |

# PDF Task 映射 • 019–027 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 29 | LUNA-018 | LUNA | Product Gate / App Gate 可重复绿化 | R15 | PARTIAL | 让当前 main 在手动/PR Gate 上真实执行并绿；红色不得靠 continue-on-error、跳测试或改成 advisory 消失。 | Tier A/B clean checkout green；App analyze/test green；latest run 对 current SHA 有证据。 |
| 30 | LUNA-019 | LUNA | Legacy 名称 / 兼容标识逐项清理 | R02 | PARTIAL | 清理用户可见与无必要的历史命名；对兼容 ID/mutex/env 逐项分类，不做全仓字符串替换。 | 用户可见只剩 序境/LUMENIS/BMP/CTH；保留兼容标识有注释/owner/迁移理由；3 个显著 legacy path 有处置。 |
| 31 | SOL-020 | SOL | Experience 共用 Primitive 边界审核 | R08 | PARTIAL | 决定哪些行为可成为共享 primitive（typography/input/evidence/state/focus/motion），哪些必须留在 screen composition，防止 Luna 做 universal renderer。 | 输出共享 primitive whitelist + screen-specific composition red lines。 |
| 32 | LUNA-021 | LUNA | Flutter Shell Context-aware 重构 | R08 | PARTIAL | 把现有五目的地工程壳重构成 context-aware workspace：Subject/Brand/Task/Runtime 状态清晰，但不把 44 个 route 平铺成导航。 | 360/390/430/768/1024/1280/1440/1920 无 overflow；navigation/restoration/context 可测试。 |
| 33 | LUNA-022 | LUNA | State / Evidence / Focus 共享 Primitive | R08 | PARTIAL | 实现可复用但不决定页面布局的 State、Evidence anchor、Trace/Return、focus、loading/error/offline 等语义 primitives。 | 各 screen 可组合使用；无视觉骨架绑定；键盘/语义/屏读可测试。 |
| 34 | LUNA-023 | LUNA | 三语 Copy Key / 长文本 / Locale 基础 | R08 | PARTIAL | 把 machine ID→semantic key→ZH/EN/KO display 的基础接到 Flutter；真实 business data 不进 locale copy。 | 关键 shell/state/action copy 可三语；long text 不溢出；fallback 明确。 |
| 35 | SOL-024 | SOL | Golden Pages 开工顺序与验收门 | R09 | PARTIAL | 确认五 Golden Pages + CTH-02 的实现顺序，要求先关闭对应 backend scope，再做 UI，再做 acceptance。 | 每个 Golden Page 有独立 GO 条件；后续 remaining screens 才能按 workflow family 展开。 |
| 36 | SOL-025 | SOL | ACC-01 \| 认证入口：Screen Spec 闭合 | R14 | ABSENT | 把 ACC-01 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：安静的身份入口；仅展示真实支持的认证方式，内测前不得形成全局产品账号墙。；主操作：凭据/已支持登录 → 校验/拒绝/恢复；业务读取仍需后续 Permission。；移动：单列表单，显式返回与错误恢复。；所 有字段/动作有 source 或明确 BLOCKED。 |
| 37 | LUNA-026 | LUNA | ACC-01 \| 认证入口：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 ACC-01 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 保留 Core Basic 与 v4 account 两条现有身份通道；先验证真实行为，再绑定逐屏权限。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |

# PDF Task 映射 • 028–036 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 38 | LUNA-027 | LUNA | ACC-01 \| 认证入口：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 ACC-01 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：安静的身份入口；仅展示真实支持的认证方式，内测前不得形成全局产品账号墙。；主交互完成：凭 据/已支持登录 → 校验/拒绝/恢复；业务读取仍需后续 Permission。；mobile 重组：单列表单，显式返回与错误恢复。； 状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 39 | LUNA-028 | LUNA | ACC-01 \| 认证入口：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 ACC-01 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以 “页面能渲染” 作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 40 | SOL-029 | SOL | ACC-02 \| 主体与 Lens 选择：Screen Spec 闭合 | R14 | ABSENT | 把 ACC-02 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：先选择授权 Subject，再选择 CTH/BMP Lens；Subject 身份与观察 Lens 分离。；主操作：搜索/选 Subject → 选择可用 Lens → 服务端决定目的地。；移动：全屏主体列表 → 主体确认 → Lens。；所有字段/ 动作有 source 或明确 BLOCKED。 |
| 41 | LUNA-030 | LUNA | ACC-02 \| 主体与 Lens 选择：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 ACC-02 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 保留 selection-only registry；活动 Subject 切换必须走既有 context + per-Subject Permission。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 42 | LUNA-031 | LUNA | ACC-02 \| 主体与 Lens 选择：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 ACC-02 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：先选择授权 Subject，再选择 CTH/BMP Lens；Subject 身份与观察 Lens 分离。；主交互完成：搜索 /选 Subject → 选择可用 Lens → 服务端决定目的地。；mobile 重组：全屏主体列表 → 主体确认 → Lens。；状态/Trac e/Return/keyboard/touch/semantics 完整。 |
| 43 | LUNA-032 | LUNA | ACC-02 \| 主体与 Lens 选择：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 ACC-02 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以 “页面能渲染” 作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 44 | SOL-033 | SOL | ACC-03 \| 工作区启动：Screen Spec 闭合 | R14 | ABSENT | 把 ACC-03 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：最近工作、CTH/BMP 入口、AI 单一意图入口组成启动空间；健康状态只是小指示。；主操作：恢复可恢复上下文或开始一个明确任务。；移动：紧凑目的地 + 一个主要 Resume/Start。；所有字段/动作有 source 或明确 BLOCKED。 |
| 45 | LUNA-034 | LUNA | ACC-03 \| 工作区启动：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 ACC-03 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 将 runtime health 读绑定可信 SYSTEM actor；不得把健康状态做成业务 Dashboard。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 46 | LUNA-035 | LUNA | ACC-03 \| 工作区启动：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 ACC-03 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：最近工作、CTH/BMP 入口、AI 单一意图入口组成启动空间；健康状态只是小指示。；主交互完成：恢 复可恢复上下文或开始一个明确任务。；mobile 重组：紧凑目的地 + 一个主要 Resume/Start。；状态/Trace/Return/keybo ard/touch/semantics 完整。 |

# PDF Task 映射 • 037–045 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 47 | LUNA-036 | LUNA | ACC-03 \| 工作区启动：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 ACC-03 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 48 | SOL-037 | SOL | ACC-04 \| 访问恢复与守卫：Screen Spec 闭合 | R14 | ABSENT | 把 ACC-04 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：解释具体阻塞步骤、缺失 scope 与真实恢复路径，不伪造角色或权限。；主操作：重试 / 切换已授权上下文 / 真实 request-access（若存在）。；移动：顺序恢复步骤 + 清晰返回。；所有字段/动作有 source 或明确 BLOCKED。 |
| 49 | LUNA-038 | LUNA | ACC-04 \| 访问恢复与守卫：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 ACC-04 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 auth_recovery_center；连接可信 actor、记录 scope，并验证恢复动作边界。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 50 | LUNA-039 | LUNA | ACC-04 \| 访问恢复与守卫：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 ACC-04 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：解释具体阻塞步骤、缺失 scope 与真实恢复路径，不伪造角色或权限。；主交互完成：重试 / 切换 已授权上下文 / 真实 request-access（若存在）。；mobile 重组：顺序恢复步骤 + 清晰返回。；状态/Trace/Return/keyb oard/touch/semantics 完整。 |
| 51 | LUNA-040 | LUNA | ACC-04 \| 访问恢复与守卫：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 ACC-04 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 52 | SOL-041 | SOL | CTH-01 \| 商业运营总览：Screen Spec 闭合 | R09 | PARTIAL | 把 CTH-01 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：Subject 锚定运营台：指标脊柱、时间信号、Event/Evidence 焦点、Decision/Action 入口。；主操作：选择时间变化 → 看来源 → 进入 Case/Decision → Trace/Return。；移动：一个活动信号 + Evidence 抽 屉 + 上下文 Action。；所有字段/动作有 source 或明确 BLOCKED。 |
| 53 | LUNA-042 | LUNA | CTH-01 \| 商业运营总览：Backend / Binding 闭合 | R09 | PARTIAL | 仅关闭 CTH-01 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 给 KPI/Event/Insight 绑定可信 Subject、freshness 与 Evidence；外部客流/销售缺失保持 NOT_COLLECTED。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 54 | LUNA-043 | LUNA | CTH-01 \| 商业运营总览：App UI / UX 实现 | R09 | PARTIAL | 在 backend contract 已可信后，实现 CTH-01 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：Subject 锚定运营台：指标脊柱、时间信号、Event/Evidence 焦点、Decision/Action 入口。；主交 互完成：选择时间变化 → 看来源 → 进入 Case/Decision → Trace/Return。；mobile 重组：一个活动信号 + Evidence 抽屉 + 上下文 Action。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 55 | LUNA-044 | LUNA | CTH-01 \| 商业运营总览：Acceptance / E2E 证据 | R09 | PARTIAL | 独立验证 CTH-01 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |

# PDF Task 映射 • 046–054 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 56 | SOL-045 | SOL | CTH-02 \| 经营主体与关联信息：Screen Spec 闭合 | R09 | PARTIAL | 把 CTH-02 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：Subject → 业务单元 → 品牌/业态的关系层级；目录选择与业务授权分离。；主操作：展开关系看 provenance；切换 Subject 只能走授权 context control。；移动：面包屑层级 + 单关系焦点。；所有字段/动 作有 source 或明确 BLOCKED。 |
| 57 | LUNA-046 | LUNA | CTH-02 \| 经营主体与关联信息：Backend / Binding 闭合 | R09 | PARTIAL | 仅关闭 CTH-02 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 当前仅 selection-only directory 可用；完整 profile/关系必须找到 canonical owner + Evidence 后再开放。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/ E2E。 |
| 58 | LUNA-047 | LUNA | CTH-02 \| 经营主体与关联信息：App UI / UX 实现 | R09 | PARTIAL | 在 backend contract 已可信后，实现 CTH-02 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：Subject → 业务单元 → 品牌/业态的关系层级；目录选择与业务授权分离。；主交互完成：展开关 系看 provenance；切换 Subject 只能走授权 context control。；mobile 重组：面包屑层级 + 单关系焦点。；状态/Trace /Return/keyboard/touch/semantics 完整。 |
| 59 | LUNA-048 | LUNA | CTH-02 \| 经营主体与关联信息：Acceptance / E2E 证据 | R09 | PARTIAL | 独立验证 CTH-02 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 60 | SOL-049 | SOL | CTH-03 \| 实时监测：Screen Spec 闭合 | R10 | ABSENT | 把 CTH-03 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：以 freshness 时钟、实时/已确认区分、source lanes 与 collector health 构成实时节奏。；主操作：暂停/恢复视图，pin Event，打开 source Evidence。；移动：优先级单流 + 持久 freshness/connection。 ；所有字段/动作有 source 或明确 BLOCKED。 |
| 61 | LUNA-050 | LUNA | CTH-03 \| 实时监测：Backend / Binding 闭合 | R10 | PARTIAL | 仅关闭 CTH-03 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 verified notes/negative source；补 Subject scope、filter semantics、Evidence link。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 62 | LUNA-051 | LUNA | CTH-03 \| 实时监测：App UI / UX 实现 | R10 | ABSENT | 在 backend contract 已可信后，实现 CTH-03 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：以 freshness 时钟、实时/已确认区分、source lanes 与 collector health 构成实时节奏。；主交 互完成：暂停/恢复视图，pin Event，打开 source Evidence。；mobile 重组：优先级单流 + 持久 freshness/connection 。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 63 | LUNA-052 | LUNA | CTH-03 \| 实时监测：Acceptance / E2E 证据 | R10 | ABSENT | 独立验证 CTH-03 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 64 | SOL-053 | SOL | CTH-04 \| 事件关系：Screen Spec 闭合 | R10 | ABSENT | 把 CTH-04 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：事件因果星图 + 时间轴；节点表达 Event/Case/Evidence 关系，而不是卡片列表。；主操作：选节点 → Evidence/Case → Return 回同一图位置。；移动：纵向事件路径 + Evidence 节点，保存 zoom/focus。； 所有字段/动作有 source 或明确 BLOCKED。 |

# PDF Task 映射 • 055–063 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 65 | LUNA-054 | LUNA | CTH-04 \| 事件关系：Backend / Binding 闭合 | R10 | PARTIAL | 仅关闭 CTH-04 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 当前事件存在但真实关系 edge 未供应；先验证 canonical edge owner 与 Evidence。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 66 | LUNA-055 | LUNA | CTH-04 \| 事件关系：App UI / UX 实现 | R10 | ABSENT | 在 backend contract 已可信后，实现 CTH-04 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：事件因果星图 + 时间轴；节点表达 Event/Case/Evidence 关系，而不是卡片列表。；主交互完成： 选节点 → Evidence/Case → Return 回同一图位置。；mobile 重组：纵向事件路径 + Evidence 节点，保存 zoom/focus。 ；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 67 | LUNA-056 | LUNA | CTH-04 \| 事件关系：Acceptance / E2E 证据 | R10 | ABSENT | 独立验证 CTH-04 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 68 | SOL-057 | SOL | CTH-05 \| 情绪日历：Screen Spec 闭合 | R10 | ABSENT | 把 CTH-05 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：日历热场表达日期的事件/情绪密度，并明确数据可用性与色阶。；主操作：选日/区间 → 事件与来源；键盘可切 list view。；移动：周/日 agenda + 密度标记 + Evidence 列表。；所有字段/动作有 source 或 明确 BLOCKED。 |
| 69 | LUNA-058 | LUNA | CTH-05 \| 情绪日历：Backend / Binding 闭合 | R10 | PARTIAL | 仅关闭 CTH-05 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 保留 daily source；修复 read error → 0，逐日绑定 Subject/source record。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 70 | LUNA-059 | LUNA | CTH-05 \| 情绪日历：App UI / UX 实现 | R10 | ABSENT | 在 backend contract 已可信后，实现 CTH-05 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：日历热场表达日期的事件/情绪密度，并明确数据可用性与色阶。；主交互完成：选日/区间 → 事件 与来源；键盘可切 list view。；mobile 重组：周/日 agenda + 密度标记 + Evidence 列表。；状态/Trace/Return/keyboa rd/touch/semantics 完整。 |
| 71 | LUNA-060 | LUNA | CTH-05 \| 情绪日历：Acceptance / E2E 证据 | R10 | ABSENT | 独立验证 CTH-05 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 72 | SOL-061 | SOL | CTH-06 \| 负面监测与反馈：Screen Spec 闭合 | R10 | ABSENT | 把 CTH-06 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：将舆情、用户反馈、已确认事件分流为 triage queue；source-first 阅读。；主操作：核验 Source → 分类/关联 Case → 保留原始措辞与 lineage。；移动：Review 卡片栈 + 全屏 source reader。；所有字段 /动作有 source 或 明确 BLOCKED。 |
| 73 | LUNA-062 | LUNA | CTH-06 \| 负面监测与反馈：Backend / Binding 闭合 | R10 | PARTIAL | 仅关闭 CTH-06 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 映射真实 note 字段，不得把 note 重命名成 Case；Case/Evidence 只走 canonical owner。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |

# PDF Task 映射 • 064–072 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 74 | LUNA-063 | LUNA | CTH-06 \| 负面监测与反馈：App UI / UX 实现 | R10 | ABSENT | 在 backend contract 已可信后，实现 CTH-06 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：将舆情、用户反馈、已确认事件分流为 triage queue；source-first 阅读。；主交互完成：核验 So urce → 分类/关联 Case → 保留原始措辞与 lineage。；mobile 重组：Review 卡片栈 + 全屏 source reader。；状态/Tr ace/Return/keyboard/touch/semantics 完整。 |
| 75 | LUNA-064 | LUNA | CTH-06 \| 负面监测与反馈：Acceptance / E2E 证据 | R10 | ABSENT | 独立验证 CTH-06 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 76 | SOL-065 | SOL | CTH-07 \| 趋势分析：Screen Spec 闭合 | R10 | ABSENT | 把 CTH-07 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：同步时间序列 + metric definition + comparison overlay + delta decomposition。；主操作：添加兼容 metric/cohort → 看 delta 贡献与 Evidence。；移动：单主图 + 横向 metric switcher + delta 解释 。；所有字段/动作有 source 或明确 BLOCKED。 |
| 77 | LUNA-066 | LUNA | CTH-07 \| 趋势分析：Backend / Binding 闭合 | R10 | PARTIAL | 仅关闭 CTH-07 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 修复 daily aggregate 的 error-to-zero；补 Subject/freshness/Evidence。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 78 | LUNA-067 | LUNA | CTH-07 \| 趋势分析：App UI / UX 实现 | R10 | ABSENT | 在 backend contract 已可信后，实现 CTH-07 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：同步时间序列 + metric definition + comparison overlay + delta decomposition。；主交互完成：添加兼容 metric/cohort → 看 delta 贡献与 Evidence。；mobile 重组：单主图 + 横向 metric switcher + delta 解 释。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 79 | LUNA-068 | LUNA | CTH-07 \| 趋势分析：Acceptance / E2E 证据 | R10 | ABSENT | 独立验证 CTH-07 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 80 | SOL-069 | SOL | CTH-08 \| 案例与知识库：Screen Spec 闭合 | R11 | ABSENT | 把 CTH-08 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：Open Case 生命周期与 verified knowledge shelf 分开；Case timeline 不等于知识。；主操作：授权状态迁移；reviewed conclusion 才能晋升 knowledge。；移动：Case inbox + 步进 timeline。；所有字段/ 动作有 source 或明确 BLOCKED。 |
| 81 | LUNA-070 | LUNA | CTH-08 \| 案例与知识库：Backend / Binding 闭合 | R11 | PARTIAL | 仅关闭 CTH-08 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 当前 high-risk/event 不是 canonical Case；先绑定 Case owner、ID、state、lifecycle。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 82 | LUNA-071 | LUNA | CTH-08 \| 案例与知识库：App UI / UX 实现 | R11 | ABSENT | 在 backend contract 已可信后，实现 CTH-08 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：Open Case 生命周期与 verified knowledge shelf 分开；Case timeline 不等于知识。；主交互完 成：授权状态迁移；reviewed conclusion 才能晋升 knowledge。；mobile 重组：Case inbox + 步进 timeline。；状态/Tr ace/Return/keyboard/touch/semantics 完整。 |

# PDF Task 映射 • 073–081 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 83 | LUNA-072 | LUNA | CTH-08 \| 案例与知识库：Acceptance / E2E 证据 | R11 | ABSENT | 独立验证 CTH-08 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 84 | SOL-073 | SOL | CTH-09 \| 风险与决策板：Screen Spec 闭合 | R11 | ABSENT | 把 CTH-09 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：Risk register 与 Decision ledger 对齐，Evidence 与 Action owner 同行展示。；主操作：看 basis → 记录 Decision → 请求审批 → 确认 Action。；移动：风险流 + Decision detail + 明确确认。；所有 字段/动作有 source 或明确 BLOCKED。 |
| 85 | LUNA-074 | LUNA | CTH-09 \| 风险与决策板：Backend / Binding 闭合 | R11 | PARTIAL | 仅关闭 CTH-09 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | Risk 可复用但 Decision owner/API 当前缺失；不得自行创造第二套 Decision。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 86 | LUNA-075 | LUNA | CTH-09 \| 风险与决策板：App UI / UX 实现 | R11 | ABSENT | 在 backend contract 已可信后，实现 CTH-09 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：Risk register 与 Decision ledger 对齐，Evidence 与 Action owner 同行展示。；主交互完成： 看 basis → 记录 Decision → 请求审批 → 确认 Action。；mobile 重组：风险流 + Decision detail + 明确确认。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 87 | LUNA-076 | LUNA | CTH-09 \| 风险与决策板：Acceptance / E2E 证据 | R11 | ABSENT | 独立验证 CTH-09 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 88 | SOL-077 | SOL | CTH-10 \| 报告与周报：Screen Spec 闭合 | R11 | ABSENT | 把 CTH-10 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：Report storyboard：结论、上下文、drivers、Evidence、方法与 scope；不是 Dashboard 截图。；主操作：选 period/block → 核验 source → preview/export。；移动：章节列表 → 单页预览 → export sheet。；所有字段/动作有 source 或明确 BLOCKED。 |
| 89 | LUNA-078 | LUNA | CTH-10 \| 报告与周报：Backend / Binding 闭合 | R11 | PARTIAL | 仅关闭 CTH-10 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 当前 report/Evidence 跨 scope；先在现有 owner 加 Subject/product filters 和 export Permission。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 90 | LUNA-079 | LUNA | CTH-10 \| 报告与周报：App UI / UX 实现 | R11 | ABSENT | 在 backend contract 已可信后，实现 CTH-10 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：Report storyboard：结论、上下文、drivers、Evidence、方法与 scope；不是 Dashboard 截图。； 主交互完成：选 period/block → 核验 source → preview/export。；mobile 重组：章节列表 → 单页预览 → export sheet。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 91 | LUNA-080 | LUNA | CTH-10 \| 报告与周报：Acceptance / E2E 证据 | R11 | ABSENT | 独立验证 CTH-10 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |

# PDF Task 映射 • 082–090 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 92 | SOL-081 | SOL | BMP-01 | 品牌市场总览：Screen Spec 闭合 | R09 | PARTIAL | 把 BMP-01 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：品牌中心关系场：audience/creator/channel/competitor 作为市场作用力，不能复用 CTH 时间线。；主操作：聚焦 BrandFocus → 看关系 Evidence → 进入 market/propagation/campaign。；移动：brand pulse + 关系轮播 + source trace。；所有字段/动作有 source 或明确 BLOCKED。 |
| 93 | LUNA-082 | LUNA | BMP-01 | 品牌市场总览：Backend / Binding 闭合 | R09 | PARTIAL | 仅关闭 BMP-01 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 必须先得到可信 actor + authorized BrandFocus + Evidence lineage，产品级 ZBMP 聚合不能当选中品牌。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E 。 |
| 94 | LUNA-083 | LUNA | BMP-01 | 品牌市场总览：App UI / UX 实现 | R09 | PARTIAL | 在 backend contract 已可信后，实现 BMP-01 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：品牌中心关系场：audience/creator/channel/competitor 作为市场作用力，不能复用 CTH 时间线。 ；主交互完成：聚焦 BrandFocus → 看关系 Evidence → 进入 market/propagation/campaign。；mobile 重组：brand pulse + 关系轮播 + source trace。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 95 | LUNA-084 | LUNA | BMP-01 | 品牌市场总览：Acceptance / E2E 证据 | R09 | PARTIAL | 独立验证 BMP-01 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以 “页面能渲染” 作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 96 | SOL-085 | SOL | BMP-02 | 品牌组合与竞品：Screen Spec 闭合 | R12 | ABSENT | 把 BMP-02 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：Owned brand 与 competitor 的 portfolio constellation；类别与身份围绕品牌。；主操作：比较/聚焦品牌；context switch 显式并可 Return。；移动：搜索 portfolio → brand profile → competitor chip s。；所有字段/动作有 source 或明确 BLOCKED。 |
| 97 | LUNA-086 | LUNA | BMP-02 | 品牌组合与竞品：Backend / Binding 闭合 | R12 | PARTIAL | 仅关闭 BMP-02 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 curated directory；真实品牌指标必须来自 BrandFocus scoped canonical BMP source。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 98 | LUNA-087 | LUNA | BMP-02 | 品牌组合与竞品：App UI / UX 实现 | R12 | ABSENT | 在 backend contract 已可信后，实现 BMP-02 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：Owned brand 与 competitor 的 portfolio constellation；类别与身份围绕品牌。；主交互完成： 比较/聚焦品牌；context switch 显式并可 Return。；mobile 重组：搜索 portfolio → brand profile → competitor chips。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 99 | LUNA-088 | LUNA | BMP-02 | 品牌组合与竞品：Acceptance / E2E 证据 | R12 | ABSENT | 独立验证 BMP-02 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以 “页面能渲染” 作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 100 | SOL-089 | SOL | BMP-03 | 市场情报：Screen Spec 闭合 | R12 | ABSENT | 把 BMP-03 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：机会雷达按主题 momentum 与 evidence confidence 组织；选中信号旁是 source brief。；主操作：过滤市场窗口 → 看 provenance → 标记待复核 lead。；移动：机会 feed + source reader。；所有字段/动 作有 source 或明确 BLOCKED。 |

# PDF Task 映射 • 091–099 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 101 | LUNA-090 | LUNA | BMP-03 \| 市场情报：Backend / Binding 闭合 | R12 | PARTIAL | 仅关闭 BMP-03 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 当前 intelligence_assets 跨 scope；在现有 owner 过滤 domain/brand 并附 item Evidence。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 102 | LUNA-091 | LUNA | BMP-03 \| 市场情报：App UI / UX 实现 | R12 | ABSENT | 在 backend contract 已可信后，实现 BMP-03 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：机会雷达按主题 momentum 与 evidence confidence 组织；选中信号旁是 source brief。；主交互 完成：过滤市场窗口 → 看 provenance → 标记待复核 lead。；mobile 重组：机会 feed + source reader。；状态/Trace /Return/keyboard/touch/semantics 完整。 |
| 103 | LUNA-092 | LUNA | BMP-03 \| 市场情报：Acceptance / E2E 证据 | R12 | ABSENT | 独立验证 BMP-03 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 104 | SOL-093 | SOL | BMP-04 \| 媒体传播分析：Screen Spec 闭合 | R12 | ABSENT | 把 BMP-04 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：channel → content → exposure → response 的传播河流与统一时间窗。；主操作：追踪 campaign/content → 看每个平台 source、metric 定义、freshness。；移动：纵向 funnel + channel selector + metric/source detail。；所有字段/动作有 source 或明确 BLOCKED。 |
| 105 | LUNA-094 | LUNA | BMP-04 \| 媒体传播分析：Backend / Binding 闭合 | R12 | PARTIAL | 仅关闭 BMP-04 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 现有 brand_propagation 读无 BrandFocus 且 GET 可有 schema side effect；先做 side-effect-free scoped read。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 106 | LUNA-095 | LUNA | BMP-04 \| 媒体传播分析：App UI / UX 实现 | R12 | ABSENT | 在 backend contract 已可信后，实现 BMP-04 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：channel → content → exposure → response 的传播河流与统一时间窗。；主交互完成：追踪 campaign/content → 看每个平台 source、metric 定义、freshness。；mobile 重组：纵向 funnel + channel selector + metric/source detail。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 107 | LUNA-096 | LUNA | BMP-04 \| 媒体传播分析：Acceptance / E2E 证据 | R12 | ABSENT | 独立验证 BMP-04 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 108 | SOL-097 | SOL | BMP-05 \| 创作者生态：Screen Spec 闭合 | R12 | ABSENT | 把 BMP-05 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：KOS/KOL/KOC/UGC creator graph，按 audience fit、真实性、贡献、风险；辅以审计表。；主操作：打开 creator dossier → 比较贡献 → 记录风险/加入授权 campaign。；移动：creator cards + dossier detail。；所有字段/动作有 source 或明确 BLOCKED。 |
| 109 | LUNA-098 | LUNA | BMP-05 \| 创作者生态：Backend / Binding 闭合 | R12 | PARTIAL | 仅关闭 BMP-05 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 profile；加 BrandFocus scope 与 canonical content/conversion/Evidence link。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |

# PDF Task 映射 • 100–108 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 110 | LUNA-099 | LUNA | BMP-05 | 创作者生态：App UI / UX 实现 | R12 | ABSENT | 在 backend contract 已可信后，实现 BMP-05 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：KOS/KOL/KOC/UGC creator graph，按 audience fit 、真实性、贡献、风险；辅以审计表。；主交互 完成：打开 creator dossier → 比较贡献 → 记录风险/加入授权 campaign。；mobile 重组：creator cards + dossier d etail。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 111 | LUNA-100 | LUNA | BMP-05 | 创作者生态：Acceptance / E2E 证据 | R12 | ABSENT | 独立验证 BMP-05 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 112 | SOL-101 | SOL | BMP-06 | 内容中心：Screen Spec 闭合 | R13 | ABSENT | 把 BMP-06 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：Brief/Draft/Review/Approved/Scheduled/Delivered 的编辑生产板；版本绑定 rights/provenance。；主操作：上传/版本化 → Review lineage/approval → 请求 publish handoff。；移动：生产队列 + versio n stack + approval actions。；所有字段/动作有 source 或明确 BLOCKED。 |
| 113 | LUNA-102 | LUNA | BMP-06 | 内容中心：Backend / Binding 闭合 | R13 | PARTIAL | 仅关闭 BMP-06 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 当前 content Evidence 全局；先通过 Evidence owner 做 scope，错误不能假装 empty。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 114 | LUNA-103 | LUNA | BMP-06 | 内容中心：App UI / UX 实现 | R13 | ABSENT | 在 backend contract 已可信后，实现 BMP-06 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：Brief/Draft/Review/Approved/Scheduled/Delivered 的编辑生产板；版本绑定 rights/provenance 。；主交互完成：上传/版本化 → Review lineage/approval → 请求 publish handoff。；mobile 重组：生产队列 + vers ion stack + approval actions。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 115 | LUNA-104 | LUNA | BMP-06 | 内容中心：Acceptance / E2E 证据 | R13 | ABSENT | 独立验证 BMP-06 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 116 | SOL-105 | SOL | BMP-07 | Campaign 工作区：Screen Spec 闭合 | R13 | ABSENT | 把 BMP-07 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：objective → audience → channel mix → milestones → outcome；Goal 与 Actual 两条轨分开。；主操作：建/改计划 → owner/date → approval → delivery → evaluation。；移动：引导式 milestones + 紧凑 objective/status。；所有字段/动作有 source 或明确 BLOCKED。 |
| 117 | LUNA-106 | LUNA | BMP-07 | Campaign 工作区：Backend / Binding 闭合 | R13 | PARTIAL | 仅关闭 BMP-07 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 当前仅 quota/cooldown 过程状态；goals/timeline/ROI/Evidence 需 canonical campaign source。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 118 | LUNA-107 | LUNA | BMP-07 | Campaign 工作区：App UI / UX 实现 | R13 | ABSENT | 在 backend contract 已可信后，实现 BMP-07 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：objective → audience → channel mix → milestones → outcome；Goal 与 Actual 两条轨分开 。；主交互完成：建/改计划 → owner/date → approval → delivery → evaluation。；mobile 重组：引导式 milestone s + 紧凑 objective/status。；状态/Trace/Return/keyboard/touch/semantics 完整。 |

# PDF Task 映射 • 109–117 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 119 | LUNA-108 | LUNA | BMP-07 \| Campaign 工作区：Acceptance / E2E 证据 | R13 | ABSENT | 独立验证 BMP-07 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 120 | SOL-109 | SOL | BMP-08 \| 播客工作室：Screen Spec 闭合 | R13 | ABSENT | 把 BMP-08 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：音频优先：waveform + timecoded script + research source + speaker cues。；主操作：从 approved brief/Evidence 起稿 → 对齐 transcript/audio → claims review → versioned export。；移动：tran script-first + scrub controls + source drawer。；所有字段/动作有 source 或明确 BLOCKED。 |
| 121 | LUNA-110 | LUNA | BMP-08 \| 播客工作室：Backend / Binding 闭合 | R13 | PARTIAL | 仅关闭 BMP-08 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 podcast owner；去 GET schema writes，加 BrandFocus filter，保留 Evidence IDs。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 122 | LUNA-111 | LUNA | BMP-08 \| 播客工作室：App UI / UX 实现 | R13 | ABSENT | 在 backend contract 已可信后，实现 BMP-08 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：音频优先：waveform + timecoded script + research source + speaker cues。；主交互完成：从 approved brief/Evidence 起稿 → 对齐 transcript/audio → claims review → versioned export。；mobile 重组：tra nscript-first + scrub controls + source drawer。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 123 | LUNA-112 | LUNA | BMP-08 \| 播客工作室：Acceptance / E2E 证据 | R13 | ABSENT | 独立验证 BMP-08 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 124 | SOL-113 | SOL | BMP-09 \| 品牌评估：Screen Spec 闭合 | R13 | ABSENT | 把 BMP-09 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：维度 scorecard + period movement + confidence + Evidence coverage，叙事结论在 measured values 之后。；主操作：选品牌/period → 看 definition/Evidence → 只比较 aligned baseline。；移动：维 度优先级列表 + 单选 Evidence trend。；所有字段/动作有 source 或明确 BLOCKED。 |
| 125 | LUNA-114 | LUNA | BMP-09 \| 品牌评估：Backend / Binding 闭合 | R13 | PARTIAL | 仅关闭 BMP-09 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 evaluation owner；read init 安全、brand scope、Evidence ref 不可丢。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 126 | LUNA-115 | LUNA | BMP-09 \| 品牌评估：App UI / UX 实现 | R13 | ABSENT | 在 backend contract 已可信后，实现 BMP-09 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：维度 scorecard + period movement + confidence + Evidence coverage，叙事结论在 measured va lues 之后。；主交互完成：选品牌/period → 看 definition/Evidence → 只比较 aligned baseline。；mobile 重组：维 度优先级列表 + 单选 Evidence trend。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 127 | LUNA-116 | LUNA | BMP-09 \| 品牌评估：Acceptance / E2E 证据 | R13 | ABSENT | 独立验证 BMP-09 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |

# PDF Task 映射 • 118–126 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 128 | SOL-117 | SOL | BMP-10 \| 品牌风险与事件：Screen Spec 闭合 | R13 | ABSENT | 把 BMP-10 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：按 severity/spread/recency 组织品牌 incident；active event timeline 连接 response owner。；主操作：triage → assign response → approval/effect → revisit impact。；移动：urgent queue + incident detail + action ladder。；所有字段/动作有 source 或明确 BLOCKED。 |
| 129 | LUNA-118 | LUNA | BMP-10 \| 品牌风险与事件：Backend / Binding 闭合 | R13 | PARTIAL | 仅关闭 BMP-10 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 当前错误复用 CTH Subject 风险；必须切回 BMP canonical Brand/Market risk owner。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 130 | LUNA-119 | LUNA | BMP-10 \| 品牌风险与事件：App UI / UX 实现 | R13 | ABSENT | 在 backend contract 已可信后，实现 BMP-10 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：按 severity/spread/recency 组织品牌 incident；active event timeline 连接 response owner。；主交互完成：triage → assign response → approval/effect → revisit impact。；mobile 重组：urgent queue + incident detail + action ladder。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 131 | LUNA-120 | LUNA | BMP-10 \| 品牌风险与事件：Acceptance / E2E 证据 | R13 | ABSENT | 独立验证 BMP-10 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 132 | SOL-121 | SOL | BMP-11 \| Benchmark 与竞品分析：Screen Spec 闭合 | R13 | ABSENT | 把 BMP-11 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：normalized comparison workbench：定义、并排品牌/竞品、Evidence strip，不可比值直接 withheld。；主操作：选择 comparison set/window → 看 normalization → supported gap → export。；移动：可 滑动比较 + sticky metric definition。；所有字段/动作有 source 或明确 BLOCKED。 |
| 133 | LUNA-122 | LUNA | BMP-11 \| Benchmark 与竞品分析：Backend / Binding 闭合 | R13 | PARTIAL | 仅关闭 BMP-11 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 benchmark owner；scope 到 selected Brand/market + Evidence 后才显示 ranking。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 134 | LUNA-123 | LUNA | BMP-11 \| Benchmark 与竞品分析：App UI / UX 实现 | R13 | ABSENT | 在 backend contract 已可信后，实现 BMP-11 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：normalized comparison workbench：定义、并排品牌/竞品、Evidence strip，不可比值直接 withheld。；主交互完成：选择 comparison set/window → 看 normalization → supported gap → export。；mobile 重组：可 滑动比较 + sticky metric definition。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 135 | LUNA-124 | LUNA | BMP-11 \| Benchmark 与竞品分析：Acceptance / E2E 证据 | R13 | ABSENT | 独立验证 BMP-11 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 136 | SOL-125 | SOL | AI-01 \| AI 意图入口：Screen Spec 闭合 | R09 | PARTIAL | 把 AI-01 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：单一 intent composer + 当前产品上下文/Evidence preview；建议任务必须来自可用能力。；主操作：输入 task → 确认 context/plan → executable 时才 start。；移动：单 prompt + context chip + expandable plan。；所有字段/动作有 source 或明确 BLOCKED。 |

# PDF Task 映射 • 127–135 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 137 | LUNA-126 | LUNA | AI-01 \| AI 意图入口：Backend / Binding 闭合 | R09 | PARTIAL | 仅关闭 AI-01 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 保留 ai_intent + PermissionGate；先绑定可信 actor/context；provider configured/idle/offline 分开。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E 。 |
| 138 | LUNA-127 | LUNA | AI-01 \| AI 意图入口：App UI / UX 实现 | R09 | PARTIAL | 在 backend contract 已可信后，实现 AI-01 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：单一 intent composer + 当前产品上下文/Evidence preview；建议任务必须来自可用能力。；主交 互完成：输入 task → 确认 context/plan → executable 时才 start。；mobile 重组：单 prompt + context chip + exp andable plan。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 139 | LUNA-128 | LUNA | AI-01 \| AI 意图入口：Acceptance / E2E 证据 | R09 | PARTIAL | 独立验证 AI-01 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 140 | SOL-129 | SOL | AI-02 \| 能力任务工作台：Screen Spec 闭合 | R14 | ABSENT | 把 AI-02 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：可取消 stages + 阶段间 artifact；每个 transition 暴露 Evidence 与 Permission 要求。；主操作：start/pause/cancel/retry safe stage；gated action 必须 review。；移动：纵向 progress path + pers istent cancel + Evidence expand。；所有字段/动作有 source 或明确 BLOCKED。 |
| 141 | LUNA-130 | LUNA | AI-02 \| 能力任务工作台：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 AI-02 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 execution owner；先增加 actor/product-scoped read，writes 仍走 effect boundary。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 142 | LUNA-131 | LUNA | AI-02 \| 能力任务工作台：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 AI-02 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：可取消 stages + 阶段间 artifact；每个 transition 暴露 Evidence 与 Permission 要求。；主交 互完成：start/pause/cancel/retry safe stage；gated action 必须 review。；mobile 重组：纵向 progress path + per sistent cancel + Evidence expand。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 143 | LUNA-132 | LUNA | AI-02 \| 能力任务工作台：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 AI-02 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 144 | SOL-133 | SOL | AI-03 \| Evidence 锚定分析线程：Screen Spec 闭合 | R14 | ABSENT | 把 AI-03 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：按结论组织 claim；每个 claim 内联 Evidence anchor、uncertainty、provenance rail。；主操作：开 source relevant passage → challenge/refine claim → follow-up task。；移动：claim cards + inli ne citations + full-screen source return。；所有字段/动作有 source 或明确 BLOCKED。 |
| 145 | LUNA-134 | LUNA | AI-03 \| Evidence 锚定分析线程：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 AI-03 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | conversation store 当前不存在；Evidence 还跨 scope。先定义 thread scope/retention，再做 execution→Evidence join。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；accept ance 仍需 UI/E2E。 |

# PDF Task 映射 • 136–144 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 146 | LUNA-135 | LUNA | AI-03 \| Evidence 锚定分析线程：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 AI-03 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：按结论组织 claim；每个 claim 内联 Evidence anchor、uncertainty、provenance rail。；主交互 完成：开 source relevant passage → challenge/refine claim → follow-up task。；mobile 重组：claim cards + inl ine citations + full-screen source return。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 147 | LUNA-136 | LUNA | AI-03 \| Evidence 锚定分析线程：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 AI-03 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 148 | SOL-137 | SOL | AI-04 \| Agent 与 Skill 目录：Screen Spec 闭合 | R14 | ABSENT | 把 AI-04 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：能力矩阵只展示 owner、任务、permission、health、last verified call；代码存在不等于可用。；主操作：按任务 filter → 看 I/O/permission → 运行 verified safe capability。；移动：可搜索 capabil ity list + detail sheet。；所有字段/动作有 source 或明确 BLOCKED。 |
| 149 | LUNA-138 | LUNA | AI-04 \| Agent 与 Skill 目录：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 AI-04 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 绑定 canonical SkillRegistry，不用旧 FUNCTION_MAP 作为最终 truth；registration 与 health 分开。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 150 | LUNA-139 | LUNA | AI-04 \| Agent 与 Skill 目录：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 AI-04 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：能力矩阵只展示 owner、任务、permission、health、last verified call；代码存在不等于可用。；主交互完成：按任务 filter → 看 I/O/permission → 运行 verified safe capability。；mobile 重组：可搜索 capab ility list + detail sheet。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 151 | LUNA-140 | LUNA | AI-04 \| Agent 与 Skill 目录：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 AI-04 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 152 | SOL-141 | SOL | AI-05 \| AI Gateway 与 Providers：Screen Spec 闭合 | R14 | ABSENT | 把 AI-05 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：Provider topology 表示 availability/latency/policy/fallback，Rules/Data 保持核心主线。；主操作：看 provider state 或 diagnostic；fallback 可观察且不阻塞核心业务。；移动：provider list + fal lback/last-check。；所有字段/动作有 source 或明确 BLOCKED。 |
| 153 | LUNA-142 | LUNA | AI-05 \| AI Gateway 与 Providers：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 AI-05 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 AI Gateway status；readiness 必须来自真实 probe，unknown/offline 保留。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 154 | LUNA-143 | LUNA | AI-05 \| AI Gateway 与 Providers：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 AI-05 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：Provider topology 表示 availability/latency/policy/fallback，Rules/Data 保持核心主线。；主交互完成：看 provider state 或 diagnostic；fallback 可观察且不阻塞核心业务。；mobile 重组：provider list + f allback/last-check。；状态/Trace/Return/keyboard/touch/semantics 完整。 |

# PDF Task 映射 • 145–153 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 155 | LUNA-144 | LUNA | AI-05 \| AI Gateway 与 Providers：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 AI-05 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 156 | SOL-145 | SOL | SYS-01 \| Runtime 健康：Screen Spec 闭合 | R09 | PARTIAL | 把 SYS-01 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：App → API → DB → Adapter 的运行 topology，显示 degradation 与 check time。；主操作：开 failing dependency diagnostic；authorized recovery 才可执行。；移动：ordered dependency list + focused diagnostic。；所有字段/动作有 source 或明确 BLOCKED。 |
| 157 | LUNA-146 | LUNA | SYS-01 \| Runtime 健康：Backend / Binding 闭合 | R09 | PARTIAL | 仅关闭 SYS-01 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 现有 RuntimeGuardian read slice 可用；补 SYSTEM actor Permission，null/offline 不能变 healthy。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 158 | LUNA-147 | LUNA | SYS-01 \| Runtime 健康：App UI / UX 实现 | R09 | PARTIAL | 在 backend contract 已可信后，实现 SYS-01 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：App → API → DB → Adapter 的运行 topology，显示 degradation 与 check time。；主交互完成：开 failing dependency diagnostic；authorized recovery 才可执行。；mobile 重组：ordered dependency list + focused diagnostic。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 159 | LUNA-148 | LUNA | SYS-01 \| Runtime 健康：Acceptance / E2E 证据 | R09 | PARTIAL | 独立验证 SYS-01 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 160 | SOL-149 | SOL | SYS-02 \| Collector 状态：Screen Spec 闭合 | R14 | ABSENT | 把 SYS-02 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：platform lanes 展示 scheduler/rate-limit/retry/backoff/queue/ingestion stage。；主操作：看 run → retry/stop（若允许）→ trace collected record。；移动：platform queue cards + running/retry 状态。；所有字段/动作有 source 或明确 BLOCKED。 |
| 161 | LUNA-150 | LUNA | SYS-02 \| Collector 状态：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SYS-02 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 collector status owner；验证 health semantics 与 operator access。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 162 | LUNA-151 | LUNA | SYS-02 \| Collector 状态：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SYS-02 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：platform lanes 展示 scheduler/rate-limit/retry/backoff/queue/ingestion stage。；主交互完成：看 run → retry/stop（若允许）→ trace collected record。；mobile 重组：platform queue cards + running/retry 状态。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 163 | LUNA-152 | LUNA | SYS-02 \| Collector 状态：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SYS-02 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |

# PDF Task 映射 • 154–162 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 164 | SOL-153 | SOL | SYS-03 | 数据库与 Truth Chain：Screen Spec 闭合 | R14 | ABSENT | 把 SYS-03 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：schema/version/integrity 分层连接 source → canonical → evidence → metric。；主操作：检查 table/chain validation → 精确 integrity error；recovery 独立授权。；移动：chain stages + integri ty summary + error detail。；所有字段/动作有 source 或明确 BLOCKED。 |
| 165 | LUNA-154 | LUNA | SYS-03 | 数据库与 Truth Chain：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SYS-03 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 canonical_data self_check/schema versions；加 SYSTEM reader auth，healthy/failed/unknown 分明。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E 2E。 |
| 166 | LUNA-155 | LUNA | SYS-03 | 数据库与 Truth Chain：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SYS-03 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：schema/version/integrity 分层连接 source → canonical → evidence → metric。；主交互完成：检查 table/chain validation → 精确 integrity error；recovery 独立授权。；mobile 重组：chain stages + integr ity summary + error detail。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 167 | LUNA-156 | LUNA | SYS-03 | 数据库与 Truth Chain：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SYS-03 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 168 | SOL-157 | SOL | SYS-04 | 安全与权限：Screen Spec 闭合 | R14 | ABSENT | 把 SYS-04 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：actor × product/domain × operation/risk 的 Permission matrix + 独立 audit trail。；主操作：模拟/读取 access decision → reason；真正 approve 只走真实二因子/审批。；移动：role/scope list + decision detail。；所有字段/动作有 source 或明确 BLOCKED。 |
| 169 | LUNA-158 | LUNA | SYS-04 | 安全与权限：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SYS-04 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 当前只是 enums，不是 Permission decision；绑定真实 permission callable。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 170 | LUNA-159 | LUNA | SYS-04 | 安全与权限：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SYS-04 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：actor × product/domain × operation/risk 的 Permission matrix + 独立 audit trail。；主交 互完成：模拟/读取 access decision → reason；真正 approve 只走真实二因子/审批。；mobile 重组：role/scope list + decision detail。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 171 | LUNA-160 | LUNA | SYS-04 | 安全与权限：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SYS-04 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 172 | SOL-161 | SOL | SYS-05 | 设置与外观：Screen Spec 闭合 | R14 | ABSENT | 把 SYS-05 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：密度/对比度/语言/motion 的 preference studio，只影响 presentation，不改 business data。；主操作：改 preference → preview → save/reset → 明确 persisted scope。；移动：分组设置页 + 实时无 障碍 preview。；所有字段/动作有 source 或明确 BLOCKED。 |

# PDF Task 映射 • 163–171 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 173 | LUNA-162 | LUNA | SYS-05 | 设置与外观：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SYS-05 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 当前仅 local preference；保持 local 或由 Sol 明确 adoption 后才引入 canonical settings persistence。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E 2E。 |
| 174 | LUNA-163 | LUNA | SYS-05 | 设置与外观：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SYS-05 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：密度/对比度/语言/motion 的 preference studio，只影响 presentation，不改 business data。；主交互完成：改 preference → preview → save/reset → 明确 persisted scope。；mobile 重组：分组设置页 + 实时无 障碍 preview。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 175 | LUNA-164 | LUNA | SYS-05 | 设置与外观：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SYS-05 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 176 | SOL-165 | SOL | SYS-06 | 内部浏览器：Screen Spec 闭合 | R14 | ABSENT | 把 SYS-06 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：受控 Evidence capture workspace：source page、capture metadata、audit action 三者分明。；主操作：navigate allowed target → pin snapshot as Evidence → inspect audit。；移动：source full-sc reen + metadata/trace panel。；所有字段/动作有 source 或明确 BLOCKED。 |
| 177 | LUNA-166 | LUNA | SYS-06 | 内部浏览器：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SYS-06 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 Browser Surface；actor/session/target 需安全授权，并保留 session provenance。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 178 | LUNA-167 | LUNA | SYS-06 | 内部浏览器：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SYS-06 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：受控 Evidence capture workspace：source page、capture metadata、audit action 三者分明。；主交互完成：navigate allowed target → pin snapshot as Evidence → inspect audit。；mobile 重组：source full-s creen + metadata/trace panel。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 179 | LUNA-168 | LUNA | SYS-06 | 内部浏览器：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SYS-06 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 180 | SOL-169 | SOL | SYS-07 | 评估与验收：Screen Spec 闭合 | R14 | ABSENT | 把 SYS-07 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：六道 acceptance gate 展示 owner/evidence/result/blocker；PARTIAL/UNKNOWN 明确未完成。；主操作：开 gate artifact → rerun permitted check → 看 exact criterion。；移动：sequential gate cards + blocker + artifact links。；所有字段/动作有 source 或明确 BLOCKED。 |
| 181 | LUNA-170 | LUNA | SYS-07 | 评估与验收：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SYS-07 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | security posture 与 runtime health 分开；绑定 current checks + authorized operator。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |

# PDF Task 映射 • 172–180 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 182 | LUNA-171 | LUNA | SYS-07 | 评估与验收：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SYS-07 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：六道 acceptance gate 展示 owner/evidence/result/blocker；PARTIAL/UNKNOWN 明确未完成。；主 交互完成：开 gate artifact → rerun permitted check → 看 exact criterion。；mobile 重组：sequential gate card s + blocker + artifact links。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 183 | LUNA-172 | LUNA | SYS-07 | 评估与验收：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SYS-07 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 184 | SOL-173 | SOL | SYS-08 | 可观测性与日志：Screen Spec 闭合 | R14 | ABSENT | 把 SYS-08 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：trace waterfall 关联 request/span/structured error；prototype events 独立 lane。；主操作：按 trace/time/component filter → 相关 logs → Return originating runtime/task。；移动：trace summa ry + spans/errors。；所有字段/动作有 source 或明确 BLOCKED。 |
| 185 | LUNA-174 | LUNA | SYS-08 | 可观测性与日志：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SYS-08 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 保留 sanitized logs；加 SYSTEM auth 与 provenance。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 186 | LUNA-175 | LUNA | SYS-08 | 可观测性与日志：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SYS-08 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操 作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：trace waterfall 关联 request/span/structured error；prototype events 独立 lane。；主交互 完成：按 trace/time/component filter → 相关 logs → Return originating runtime/task。；mobile 重组：trace sum mary + spans/errors。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 187 | LUNA-176 | LUNA | SYS-08 | 可观测性与日志：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SYS-08 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 188 | SOL-177 | SOL | SH-01 | 内容创作工作室：Screen Spec 闭合 | R09 | PARTIAL | 把 SH-01 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：Source/Brief/Evidence → editable artifact → version/outcome；Artifact 是中心。；主操作：真实 brief 起草 → cite Evidence → review → human approval handoff。；移动：focused editor + sour ce drawer + version/action sheet。；所有字段/动作有 source 或明确 BLOCKED。 |
| 189 | LUNA-178 | LUNA | SH-01 | 内容创作工作室：Backend / Binding 闭合 | R09 | PARTIAL | 仅关闭 SH-01 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 当前只安全实现 capability catalog；Studio/Pattern/Fact Wall/thread 缺口与 publish effect 需逐 owner 关闭。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍 需 UI/E2E。 |
| 190 | LUNA-179 | LUNA | SH-01 | 内容创作工作室：App UI / UX 实现 | R09 | PARTIAL | 在 backend contract 已可信后，实现 SH-01 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作 ，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：Source/Brief/Evidence → editable artifact → version/outcome；Artifact 是中心。；主交互 完成：真实 brief 起草 → cite Evidence → review → human approval handoff。；mobile 重组：focused editor + s o urce drawer + version/action sheet。；状态/Trace/Return/keyboard/touch/semantics 完整。 |

# PDF Task 映射 • 181–189 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 191 | LUNA-180 | LUNA | SH-01 \| 内容创作工作室：Acceptance / E2E 证据 | R09 | PARTIAL | 独立验证 SH-01 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 192 | SOL-181 | SOL | SH-02 \| Evidence Library：Screen Spec 闭合 | R14 | ABSENT | 把 SH-02 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：搜索优先 Evidence atlas，按 source/time/domain/lineage；snapshot 与 citation 相邻。；主操作：search/filter → 对照 source/extract → 看 freshness → 回 owning CTH/BMP。；移动：Evidence cards + full-screen source trace。；所有字段/动作有 source 或明确 BLOCKED。 |
| 193 | LUNA-182 | LUNA | SH-02 \| Evidence Library：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SH-02 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 现有 Evidence query/stats 全局；必须 actor/domain scoped 并逐 row validate。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 194 | LUNA-183 | LUNA | SH-02 \| Evidence Library：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SH-02 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：搜索优先 Evidence atlas，按 source/time/domain/lineage；snapshot 与 citation 相邻。；主交互完成：search/filter → 对照 source/extract → 看 freshness → 回 owning CTH/BMP。；mobile 重组：Evidence cards + full-screen source trace。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 195 | LUNA-184 | LUNA | SH-02 \| Evidence Library：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SH-02 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 196 | SOL-185 | SOL | SH-03 \| Canonical Data 与 Metrics 浏览：Screen Spec 闭合 | R14 | ABSENT | 把 SH-03 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：schema/metric explorer 分开 definition/owner/lineage/sample；derived 标识清楚。；主操作：选 field/metric → traverse lineage；只读，不编辑 business fact。；移动：metric list + definition + lineage drilldown。；所有字段/动作有 source 或明确 BLOCKED。 |
| 197 | LUNA-186 | LUNA | SH-03 \| Canonical Data 与 Metrics 浏览：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SH-03 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | global canonical counts 不能直接给 App；先做 authorized product/domain filtering + Evidence lineage。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/ E2E。 |
| 198 | LUNA-187 | LUNA | SH-03 \| Canonical Data 与 Metrics 浏览：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SH-03 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：schema/metric explorer 分开 definition/owner/lineage/sample；derived 标识清楚。；主交互完成：选 field/metric → traverse lineage；只读，不编辑 business fact。；mobile 重组：metric list + definition + lineage drilldown。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 199 | LUNA-188 | LUNA | SH-03 \| Canonical Data 与 Metrics 浏览：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SH-03 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |

# PDF Task 映射 • 190–198 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 200 | SOL-189 | SOL | SH-04 | 报告导出中心：Screen Spec 闭合 | R14 | ABSENT | 把 SH-04 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：configuration / preview / completed artifact / failed export 分状态；旁边展示 provenance。；主操作：选支持格式 → scope → preview → export → audited artifact。；移动：guided export + fin al artifact view。；所有字段/动作有 source 或明确 BLOCKED。 |
| 201 | LUNA-190 | LUNA | SH-04 | 报告导出中心：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SH-04 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 保留 report catalog；export 必须绑定 Permission + approval/effect fence + audit/Evidence。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 202 | LUNA-191 | LUNA | SH-04 | 报告导出中心：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SH-04 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：configuration / preview / completed artifact / failed export 分状态；旁边展示 provenance 。；主交互完成：选支持格式 → scope → preview → export → audited artifact。；mobile 重组：guided export + final artifact view。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 203 | LUNA-192 | LUNA | SH-04 | 报告导出中心：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SH-04 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 204 | SOL-193 | SOL | SH-05 | 学习与评估资产：Screen Spec 闭合 | R14 | ABSENT | 把 SH-05 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：verified SOP/retrospective/training/evaluation 与 draft 分开；provenance/review date 可见。；主操作：打开 asset → verification history → propose revision / cited task input。；移动：filte red library + reading view。；所有字段/动作有 source 或明确 BLOCKED。 |
| 205 | LUNA-194 | LUNA | SH-05 | 学习与评估资产：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SH-05 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | 复用 learning owner；加 domain/origin scope 与 lineage。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 206 | LUNA-195 | LUNA | SH-05 | 学习与评估资产：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SH-05 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：verified SOP/retrospective/training/evaluation 与 draft 分开；provenance/review date 可见 。；主交互完成：打开 asset → verification history → propose revision / cited task input。；mobile 重组：filter ed library + reading view。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 207 | LUNA-196 | LUNA | SH-05 | 学习与评估资产：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SH-05 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 208 | SOL-197 | SOL | SH-06 | 通知与任务：Screen Spec 闭合 | R14 | ABSENT | 把 SH-06 的用户问题、entry/return、PC/mobile composition、真实 fields/actions、state matrix、permission/evidence、acceptance 全部收敛成可施工 Spec。 | Spec 明确：Approval / owned task / alert / info subscription 分 urgency lanes，带 source context。；主操作：triage/assign/snooze/resolve 走授权 action；Return origin。；移动：priority queue + full-scr een task action。；所有字段/动作有 source 或明确 BLOCKED。 |

# PDF Task 映射 • 199–207 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 209 | LUNA-198 | LUNA | SH-06 | 通知与任务：Backend / Binding 闭合 | R14 | PARTIAL | 仅关闭 SH-06 依赖的真实后端 binding：复用 canonical owner，把 scope/Permission/Evidence/state 绑定完整；不做 UI。 | TaskEngine 保持 SYSTEM；operator read 要 scope，queue/offline/error 状态保真。；成功/空/未采集/离线/拒绝/错误可区分；binding registry 有当前 source evidence；acceptance 仍需 UI/E2E。 |
| 210 | LUNA-199 | LUNA | SH-06 | 通知与任务：App UI / UX 实现 | R14 | ABSENT | 在 backend contract 已可信后，实现 SH-06 的 purpose-built Flutter Experience。用户只看到任务、内容、结果与操作，系统内部 owner/tool/evidence contract 退到产品背后。 | PC 与 mobile 均表达：Approval / owned task / alert / info subscription 分 urgency lanes，带 source context。；主交互完成：triage/assign/snooze/resolve 走授权 action；Return origin。；mobile 重组：priority queue + full-sc reen task action。；状态/Trace/Return/keyboard/touch/semantics 完整。 |
| 211 | LUNA-200 | LUNA | SH-06 | 通知与任务：Acceptance / E2E 证据 | R14 | ABSENT | 独立验证 SH-06 的真实用户任务，从 App action 到 API/Permission/Canonical/Evidence/result/Return，不以“页面能渲染”作为完成。 | Widget/API/integration 至少全绿；关键 responsive widths；blocked/error/retry/Trace/Return；需要 live 的标为 LIVE_VERIFIED 或明确未验证。 |
| 212 | SOL-201 | SOL | Internal Beta Critical Journey 最终冻结 | R15 | BLOCKED | 在全部核心 Screen slice 就绪后，冻结 beta 的真实关键旅程、设备、数据源、账号/媒体 session、失败恢复与验收证据。 | 每条 journey 有 start state、真实 source、action、Evidence、result、return/evaluation、failure/recovery、device。 |
| 213 | LUNA-202 | LUNA | CTH Source→Evidence→Event→Case→Decision→Action 闭环 | R15 | BLOCKED | 用真实允许数据跑 CTH critical journey，证明 Subject scope、Evidence trace、Case/Decision/Action 与 Return/Evaluation。 | 入口到结果完整；失败/权限拒绝/离线恢复可重放；Evidence 可回源。 |
| 214 | LUNA-203 | LUNA | BMP BrandFocus→Market→Content/Creator→Campaign→Evaluation 闭环 | R15 | BLOCKED | 跑 BMP 核心旅程，证明 BrandFocus、市场/传播/创作者/内容/Campaign/评估的 scope 与 Evidence。 | BrandFocus 全程保持；每个分析可 Trace；Campaign result 与 baseline 分开；evaluation 有 lineage。 |
| 215 | LUNA-204 | LUNA | AI Intent→Task→Evidence→Artifact→Action→Evaluation 闭环 | R15 | BLOCKED | 用真实 AI Gateway/Agent runtime 跑 artifact-first 流程；不得用 chat UI 替代。 | task 有 execution id、real stage、Evidence、artifact、gated action、evaluation；provider fallback 可观察。 |
| 216 | LUNA-205 | LUNA | Shared Creation / Report / Export 闭环 | R15 | BLOCKED | 证明 brief/evidence→artifact→review/approval→export 与 lineage；publish 在无真实 publisher 时保持 blocked。 | 至少一个真实 artifact 与一个 audited export 完成；失败/blocked 状态诚实。 |
| 217 | LUNA-206 | LUNA | SYSTEM 故障→诊断→恢复→Return 闭环 | R15 | BLOCKED | 制造受控可恢复故障，证明 runtime/collector/browser/db/logs 的诊断、Permission、恢复与状态回归。 | 故障被检测、定位、恢复；用户回到原任务；audit/log 完整。 |

# PDF Task 映射 • 208–216 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 218 | LUNA-207 | LUNA | XHS/RedNote Cookie Session LIVE 验收 | R15 | BLOCKED | 在授权本地环境验证 media Cookie/session + HK collector 的真实 collection→Evidence→Canonical，不泄露凭据。 | session health、collection、provenance、Evidence、canonicalization、failure isolation 均有 live evidence。 |
| 219 | LUNA-208 | LUNA | Offline / Partial / Stale / Permission-denied 全 App 回归 | R15 | BLOCKED | 验证关键 journeys 在网络断开、部分数据、陈旧数据、403/401、collector offline 下不撒谎且可恢复。 | 关键状态均有真实 UI + semantics + recovery；Return context 保持。 |
| 220 | LUNA-209 | LUNA | Security / Permission / Effect Boundary Beta Gate | R15 | BLOCKED | 验证所有 consequential action 经过 Permission→Approval→Effect Fence，denied/pending 无副作用。 | 权限矩阵、拒绝、审批、外部 effect audit 全通过；secret scan 无泄漏。 |
| 221 | LUNA-210 | LUNA | DB Backup / Migration / Rollback Gate | R15 | BLOCKED | 用隔离 DB/production-candidate copy 验证 migration、backup、rollback、schema integrity，不修改受保护 production DB。 | migrate/rollback/restore 可重复；canonical/Evidence 数据不丢；schema owner 唯一。 |
| 222 | LUNA-211 | LUNA | Desktop 真实设备 Experience Gate | R15 | BLOCKED | 在 Windows/macOS 至少覆盖 1920/1440/1280/1024/768，跑 critical journeys、键盘/focus、resize、恢复。 | 无 overflow/blocked controls；keyboard/focus/return/long text 正常；真实 API journey 完成。 |
| 223 | LUNA-212 | LUNA | Mobile 真实设备 Experience Gate | R15 | BLOCKED | 在 Android/iOS 至少覆盖 430/390/360，验证 touch、back、interrupt/resume、offline、长文本。 | critical journey 在真实设备完成；返回/恢复正确；无横向溢出。 |
| 224 | LUNA-213 | LUNA | Accessibility + Reduced Motion Gate | R15 | BLOCKED | 全关键 screen 验证 keyboard/focus order、screen reader semantics、contrast、touch target、reduced motion。 | 关键操作无鼠标可完成；动态状态有 announcement；reduced motion 不丢信息。 |
| 225 | LUNA-214 | LUNA | Performance / Resource / Long-run Stability Gate | R15 | BLOCKED | 验证 App+Core+collector 长时间运行不出现明显内存/任务堆积/重复 supervisor/队列失控。 | 长跑期间 health 可观测；无重复 runtime；queue/retry/backoff 在界限内。 |
| 226 | LUNA-215 | LUNA | Tier A/B Clean Checkout Full Gate | R15 | BLOCKED | 在干净 clone 运行所有 blocking repository/hermetic tests 和 Flutter gate，证明不是开发机偶然通过。 | Product Gate + App Gate 对当前 commit 真实 GREEN。 |

# PDF Task 映射 • 217–225 / 225

原 Task 全保留；源码状态与多表面增量在对应 Round 重新验收。

| PDF | Task | Owner | Title | R | Source | Goal | Completion gate |
|---|---|---|---|---|---|---|---|
| 227 | LUNA-216 | LUNA | Tier C Production/Live Acceptance Gate | R15 | BLOCKED | 在满足 prerequisites 的授权环境运行 environment/live suites，补齐 A/B 无法证明的 browser/session/real data/recovery。 | 所有 beta critical prerequisites 有 dated live evidence；未跑项明确 blocker。 |
| 228 | SOL-217 | SOL | Internal Beta GO / NO-GO Review | R15 | BLOCKED | 根据 current commit 的 ACCEPTED screens、critical journeys、Tier A/B/C、security/recovery 证据决定是否进入 internal beta。 | 输出 GO 或 NO-GO + 明确 blockers；无排名式模糊语言。 |
| 229 | SOL-218 | SOL | Post-beta Account / Access Product Decision | R15 | BLOCKED | 在 internal beta 后再决定 ACC-01~04 的正式 LUMENIS account/SSO/access recovery 产品范围与 rollout。 | Account scope、SSO、roles/recovery/permission contract 被批准或明确后置。 |
| 230 | LUNA-219 | LUNA | Post-beta ACC-01~04 Production Binding / E2E | R15 | BLOCKED | 按 Sol 决策把 ACC screens 从 fail-closed/selection-only 完成到真实 product account/access flows，并跑 E2E。 | ACC flows 实际可用；授权 context 切换与恢复真实；不破坏 pre-beta verified workflows。 |
| 231 | LUNA-220 | LUNA | Compatibility Caller 最终吸收与删除 | R15 | BLOCKED | 依据 LUNA-004/SOL-005 矩阵，在所有能力已迁移/验收后删除明确 redundant compatibility callers/legacy paths。 | root compatibility 显著下降；所有删除有 replacement owner + regression；Only-One guards PASS。 |
| 232 | LUNA-221 | LUNA | Documentation / Intelligence Library 去重与权威边界 | R15 | BLOCKED | 把 docs/intelligence_library 中重复/历史材料标记为 provenance 或去重，确保只有 CURRENT Product/Experience authority 能驱动开发。 | 重复文档有 canonical pointer 或清理；legacy Z-Brand 文档不再误导当前开发。 |
| 233 | LUNA-222 | LUNA | Cross-platform Release Build Matrix | R15 | BLOCKED | 生成并验证 Windows/macOS/Linux/Android/iOS candidate build，版本/签名/配置/网络权限/启动行为一致。 | 所有目标可 build；至少目标内测平台可安装/启动/连接 Core；版本号与 artifact trace 可追。 |
| 234 | LUNA-223 | LUNA | Release Candidate Clean Clone + Fresh Data Smoke | R15 | BLOCKED | 从全新 clone 安装依赖、启动 Core/App、用新隔离/候选数据跑最短 critical smoke，排除本机残留依赖。 | fresh clone 可复现；health、App launch、一个 CTH/BMP/AI smoke 成功或真实 fail closed。 |
| 235 | SOL-224 | SOL | 最终产品闭环验收 | R15 | BLOCKED | 对 current release candidate 做最终产品级验收：Only One、App-only、44 screens 状态、critical journeys、truth/permission/evidence、平台、recovery、release evidence。 | 明确列出 ACCEPTED/DEFERRED/BLOCKED；若 release，则每个声明都能追到当前 commit + evidence。 |

APPROVED EXECUTION BASELINE

# ROUND 01 已批准；下一独立任务进入 ROUND 02

本 PDF 是覆盖与规划证据，不是设计权威、实现完成或发布声明。

## What is protected

PRESERVE

现有源码、未知本地运行态、root callers、兼容层、数据库、Evidence、Permission、Skill Store 后端资产与所有未证明已迁移的能力。

## What will happen next

SEQUENTIAL

重新 resolve 当前 SHA → 发出 ROUND 02 Charter → 仅派发第一张 bounded Luna Task Card → Sol 读取 BASE/FINAL SHA、diff、tests、runtime/E2E、blockers → PASS/REWORK/BLOCKED/SPLIT/REORDER → 停止。

## Current stop

ROUND BOUNDARY

本执行基线已获 Product Owner 批准。ROUND 02 与序境09仍未在本治理固化任务中启动；下一独立 Sol 任务必须重新解析当前 SHA、工作树、依赖与保护项。